



THE STABILITY GAP AS A TRAJECTORY PROBLEM: A BENT DESCENT PATH, ITS OVERSHOOT, AND ONE GATE IN TWO DESIGNS

Bachelor's Thesis

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Abstract: Replay-based continual learning is usually judged by the accuracy it reaches *after* each task. Continual evaluation exposes what this view hides: the *stability gap*, a sharp but transient drop in past-task accuracy at the start of every new task. The gap persists even when the optimiser descends the exact joint loss, so at least part of its cause lies in the optimisation trajectory rather than the objective. This thesis explains the gap on two levels. The *intended trajectory*, the path exact gradient descent would take, bows out of the old task's basin before returning; and the followed path overshoots the intended one through everything that vanishes with the step size: discrete updates, momentum, and gradient noise. We separate the two levels by measurement rather than by argument. Rescaling the step size is a reparametrisation of the same gradient flow, so a ladder of learning rates run at matched flow time from one frozen checkpoint discretises one and the same intended trajectory, and whatever survives $\eta \rightarrow 0$ is that trajectory. On rotated MNIST the ladder converges, and not on zero: of the 23 pp drop at the standard step size, between two thirds and three quarters is discretisation overshoot, and a smooth arc of some four points remains that neither an exact replay gradient nor any further shortening of the step removes. Reaching that limit costs a hundred times the training, which sets the bar for a method: not lowering the gap, which a shorter step does with no method at all, but lowering it at the standard step size and in the standard number of steps. Written in a unified update rule, replay methods counter both levels with a single tool, a *gate* on the new-task gradient with the replay gradient restoring at full strength, and we build and compare the two pure gate designs: an adaptive λ -curriculum (scalar, feedback) and asymmetric Fisher-preconditioned replay (directional, feedforward). The curriculum earns its keep only under momentum, and there it cuts three quarters of vanilla ER's depth at vanilla accuracy, matches in one epoch what shortening the step reaches only in the limit, and carries to CIFAR-10 on a ResNet-18 at negligible overhead. The directional gate protects the past task at no measured plasticity cost but at far higher compute, bends the arc itself on the diagnostic full-buffer leg, fails late and abruptly at its strongest setting there, and is re-inflated by momentum. Which gate is preferable is set by the momentum regime.

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1 Introduction

Deep neural networks learn powerful representations from static, independent, and identically distributed data, but a vulnerability appears as soon as training becomes sequential: the network overwrites previously acquired knowledge to accommodate new data. This phenomenon, catastrophic forgetting [19, 23], is a major hurdle in machine learning. Retraining from scratch solves the issue, but it is too expensive for systems that need to adapt continuously. Continual learning (CL) studies how to learn from a sequence of tasks without that cost, and its core difficulty is the stability–plasticity dilemma: the model must stay plastic enough to learn new distributions while staying stable enough to preserve old ones [20].

Replay, the family this thesis studies, stores a small buffer of past examples and trains on them alongside the new data. Replay methods reduce long-term forgetting well, but until recently they were evaluated only at task boundaries, which hides what happens within a task. De Lange et al. [4] evaluated the network after every update and identified the *stability gap*: a severe but transient collapse in past-task accuracy that occurs immediately when a new task begins, followed by a slow recovery. This matters in deployment: a system is queried while it trains, so a model caught mid-transition returns wrong answers on inputs its final checkpoint would handle correctly.

The discovery raised a natural question: would a better approximation of the joint loss over all tasks remove the gap? Hess et al. [12] showed that the gap persists even under incremental joint training, where the model has access to all data from all tasks at every step. A perfect objective alone does not remove transient forgetting. The problem therefore lies in *how* the optimisation proceeds, and this thesis locates it on two levels: the *intended trajectory*, the path exact gradient descent would take, already bows out of the old task’s basin; and optimisers at a step size greater than zero overshoot the trajectory they intend to follow.

Two levels. Figure 1.1 visualises the transition as a switch from a learned task (Task A) to a new task (Task B), a conceptual reduction of a high-dimensional surface using the loss-landscape lens of Kao et al. [15]. *Level one* is geometric: steepest descent on the abruptly assembled joint loss (green curve) bows out of Task A’s basin and returns from outside [15], so the Task-A loss rises above its final level and comes back, its recovery slow and lingering, the *hanging tail*. The excursion is a property of the path, not a necessary cost: Section 3.1 constructs a reference path along which the past-task loss rises monotonically to its joint-optimum level (the dashed chord), valid where its local curvature condition holds, so under that condition everything the accuracy curve does below its settled level and back is avoidable in

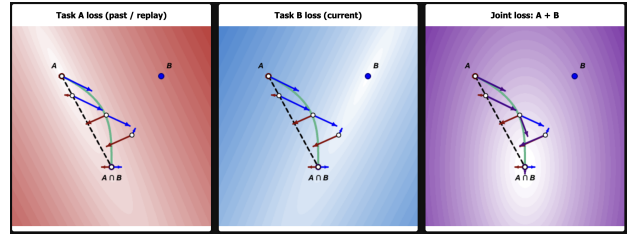


Figure 1.1: 2D optimisation landscape at a task transition. Single-task losses Task A (red) and Task B (blue) are minimised at A and B ; the joint loss (purple) is minimised at $A \cap B$. The steepest-descent trajectory (green curve) illustrates the trajectory problem (Section 3.2), bowing outside A ’s basin into regions of higher loss. In contrast, the dashed black chord sketches the monotone reference path of Section 3.1, which stays inside the basin. Overshoot drivers (Section 3.2) further push the iterate off these paths. An interactive version, in which landscapes, task minima, and both gates of Section 3.3 can be manipulated, is available at https://arthurreuss.github.io/ValleyWalk/loss_landscape_visualization.html.

principle. *Level two* is that optimisers overshoot the trajectory they intend to follow: at a step size greater than zero the large, unopposed boundary gradient [4] is applied in a few discrete jumps rather than traversed as a curve, and finite-buffer gradient noise perturbs every step [1]. Section 3 develops both levels.

One tool: gate the new-task gradient. Sequential replay methods share a single update, a weighted combination of the current-task and replay gradients [25]. Section 3.3 generalises this: because only the ratio of the two weights sets the update direction, any reweighting of this update can be rewritten with the replay gradient restoring at full strength and a single *gate* on the new-task gradient, $d = G g_{\text{cur}} + g_{\text{rep}}$. The gate’s *domain* can be scalar (attenuate the whole push) or directional (attenuate per direction), and its *control* feedback (react to a live damage signal) or feedforward (act through a schedule or a curvature model). This thesis builds and compares the two pure designs. The first is the λ -**curriculum**, a scalar gate: instead of switching abruptly to the joint loss, it ramps the new-task loss weight $\lambda(t)$ from 0 to 1, so the objective’s minimiser moves continuously from θ_A^* to θ_{joint}^* and the model tracks a moving valley floor. Its adaptive variant sets λ from the ratio of the replay and new-task gradient norms, a live damage signal, which makes it a scalar *feedback* gate. The second is **asymmetric preconditioned experience replay** (asymmetric PER), a directional *feedforward* gate: it filters the new-task gradient through the curvature of the replay loss, passing the push at full gain where the past task is flat and braking it where it is steep, leaving the restoring gradient untouched.

2 Background and Related Work

2.1 Continual Learning and Experience Replay

Methods against catastrophic forgetting fall into a few families: replay, parameter and functional regularisation, optimisation-based methods, context-dependent processing, and template-based classification [28]. This thesis focuses on replay, specifically its simplest formulation: Experience Replay (ER) [3]. At each step, ER draws a replay mini-batch from a buffer of stored past examples, combines it with the current-task mini-batch, and takes one optimiser step on the summed loss.

A standard approach to managing this episodic memory is reservoir sampling [29], which ensures that every sample seen so far occupies a buffer slot with equal probability. Under this setup, the buffer remains empty during the first task, meaning ER reduces to plain SGD; consequently, performance gaps and forgetting metrics are evaluated from the second task onward. Despite this simplicity, ER remains highly competitive: a small episodic memory often outperforms purpose-built methods and scales to longer sequences [3]. Furthermore, because the ER update consists of exactly two ingredients, it provides a remarkably clean substrate for isolating and studying interventions at task boundaries.

However, a finite buffer is inherently an imperfect estimator of the past-task distribution. Aljundi et al. [1] frame buffer selection as approximating the feasibility region of the full past data, with reservoir sampling serving as one specific policy within that framework. Consequently, the replay gradient derived from a small buffer will differ in both direction and magnitude from the true gradient of all past data.

2.2 The Stability Gap

De Lange et al. [4] identified the stability gap using continual evaluation, measuring past-task accuracy after every update rather than only at task boundaries. In this view a new task begins with substantial but transient forgetting that the model subsequently recovers from, and they quantify it with worst-case metrics, mainly the minimum past-task accuracy reached during training, rather than with end-of-task averages alone. The effect is not confined to one method: they observe it across experience replay, constraint-based replay, knowledge distillation, and parameter regularisation. It also persists well beyond the standard benchmarks, appearing in joint incremental learning of homogeneous tasks [14] and in continual pre-training of large language models [8], and several mitigations have been proposed [10, 26].

Two findings in this literature anchor the account

of Section 3. First, De Lange et al. [4] trace the initial drop to a gradient imbalance at the boundary: the new-task gradient dominates the first updates, so the optimiser trades stability for plasticity precisely when the past task is most exposed. Second, the gap is a property of the trajectory rather than the objective. Hess et al. [12] show it persists under training on the exact joint loss with full access to all past data, and Kao et al. [15] demonstrate the geometric mechanism analytically: steepest descent from an old-task minimum leaves the old basin even with perfect joint gradients, because the steepest direction and the direction to the joint minimum disagree. A perfect objective alone therefore fails to prevent the gap, and the descent geometry itself must be addressed. The optimiser matters as well: comparing optimisers in a similar setting, Obis [22] find the depth and duration of the drop to depend substantially on which one is used, an early indication that the effective step size is a lever on the gap.

2.3 One Update, Many Methods

Beyond plain ER, several methods constrain the update at the task boundary using information about past tasks. GEM stores past examples and, at every step, recomputes each past task’s gradient on its stored data, then projects the new-task gradient onto the nearest direction that leaves every stored loss unchanged or lower [17]. A-GEM replaces the per-task constraints with a single averaged past-task gradient, reducing the projection to one cheap operation [2]. EWC adds a quadratic penalty around the past optimum weighted by a curvature estimate [16], and natural-gradient approaches precondition the update with past-task curvature so that steps avoid directions the old task depends on [15].

These methods are not all of one kind: EWC and the natural-gradient family act through a penalty or a preconditioner rather than through replayed data, and they do not fit the form below. The replay-based methods, however, share a single update, as Sodhani et al. [25] show. Every step combines two gradients, the current task’s and the replay buffer’s, under two scalar weights,

$$\theta \leftarrow \theta - \eta (\alpha_1 \nabla L_{\text{current}} + \alpha_2 \nabla L_{\text{replay}}), \quad (2.1)$$

so a replay method reduces to a rule for choosing α_1 and α_2 . Plain ER fixes both to one. A-GEM holds $\alpha_1 = 1$ and raises the replay weight α_2 only when the new-task and stored gradients conflict, by just enough to cancel the interference, so its projection is equivalent to re-weighting the two losses; GEM does the same per stored task, through a quadratic program whose multi-constraint solution no longer reduces to two scalar weights. MEGA [9] likewise holds $\alpha_1 = 1$ but sets the replay weight from a loss ratio,

$\alpha_2 = L_{\text{replay}}/L_{\text{current}}$, leaning on memory once the new task is already fit.

The same reading applies to the λ -curriculum of this thesis. It fixes $\alpha_2 \equiv 1$ and turns the other knob, ramping the current-task weight $\alpha_1 = \lambda(t)$ up from zero. It belongs to the same family as A-GEM and MEGA; what distinguishes it is which weight it moves and what signal drives the weight. The adaptive variant (Section 3.3) is the mirror image of MEGA: MEGA raises the *replay* weight from a *loss* ratio to recruit memory once the new task is learned, while the adaptive curriculum holds the *new task* back with a *gradient-norm* ratio, letting it in only as the replay gradient reasserts itself.

2.4 Low-Loss Paths and Mode Connectivity

Mode connectivity gives empirical backing to the *monotone reference path* introduced in Section 3: a route from θ_A^* to θ_{joint}^* along which the past-task loss only ever rises toward its final value requires the two solutions to be joined by a corridor of low loss, and this is what the literature reports. Curved paths of near-constant low loss join independently trained minima [5, 7], straight ones join networks that share early training [6], and, for continual learning specifically, the sequentially trained and multitask solutions often lie in the same basin [21]. Where such a route exists, the stability gap reflects the optimiser wandering off it, and an intervention succeeds by keeping it there.

3 The Stability Gap as a Trajectory Problem

This section reframes the stability gap as a two-level trajectory problem. Learning a new task incurs a permanent stability-plasticity cost; the gap is the transient excess above it. This excess has two sources: a geometric bow in the *intended trajectory*, the path that exact gradient descent would follow, and the overshoot that discrete, finite-step training adds on top of it. We formalise this split by constructing a monotone reference path. We then show that gating the new-task gradient counters these deviations, though no single pure gate design neutralises every driver at once. Instead, how the update is gated dictates which source of error is suppressed and which remains exposed. Two contrasting gates instantiate this: an adaptive scalar curriculum and an asymmetric directional filter. One register note applies throughout: on the quadratic model of this section the claims are exact; carried to deep networks they are hypotheses, which Section 6 tests and Section 7 weighs.

3.1 The Reference Path and the Tracking Bound

Learning a second task inevitably moves the model. In a quadratic approximation with task losses L_A and L_B and Hessians H_A and H_B , the displacement $d = \theta_{\text{joint}}^* - \theta_A^*$ from the old optimum to the joint minimum permanently raises the replay loss by roughly $\frac{1}{2} d^\top H_A d$. Every method pays this price; it is distinct from the stability gap.

Yet this rise can be strictly monotone. Consider the curriculum objective $L_\lambda = L_{\text{replay}} + \lambda L_{\text{current}}$; at $\lambda = 1$ it is the joint loss with the two tasks weighted equally, the convention used throughout. As λ grows from 0 to 1, the minimiser traces a *reference path* $\Theta^*(\lambda)$ from θ_A^* to θ_{joint}^* . Under local positive-definiteness, an implicit-differentiation argument along the minimiser branch (Appendix D) guarantees that the replay loss climbs directly to its final level with no excursion above it:

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) \geq 0 \quad \text{for all } \lambda \in [0, 1]. \quad (3.1)$$

This guarantee is local: it holds only while the combined Hessian $H_A + \lambda H_B$ stays positive-definite along the tracked branch of minima. The condition is automatic near θ_A^* and persists until the new task’s negative curvature overpowers the old task’s; on a deep non-convex backbone it therefore serves as motivation rather than a guarantee.

Figure 3.1 shows this: as λ grows, the loss contours deform continuously from the old-task basin to the joint basin and the minimiser slides along the connected valley floor. Where the condition holds, the replay loss climbs directly to its final level, and any transient excess *above* it, the stability gap, is a preventable deviation from the reference path.

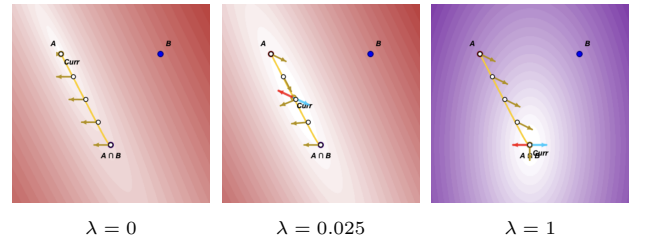


Figure 3.1: Evolution of the curriculum loss $L_\lambda = A + \lambda B$. As λ increases, the contours morph continuously from the old-task basin around A to the joint basin around $A \cap B$, and the current minimiser (*Curr*) slides along the moving valley floor, forming the reference path $\Theta^*(\lambda)$. At the minimiser the replay (red) and current-task (blue) contributions cancel by the first-order condition $\nabla L_{\text{replay}} + \lambda \nabla L_{\text{current}} = 0$. Frames from the interactive visualisation of Figure 1.1.

A real iterate cannot sit on this path exactly, but it can stay close to it. Introducing the new task gradually over an N -step ramp ($\lambda \approx 1/N$) keeps the iterate

tethered to the moving valley floor $\Theta^*(\lambda(t))$ rather than letting it leap across the landscape; a gradient-flow analysis bounds the resulting lag by $O(1/N)$ (Appendix D). Because the monotone path fixes the endpoint at the permanent level, this caps the highest transient replay loss of the whole transition,

$$\max_t L_{\text{replay}}(\theta(t)) \leq L_{\text{replay}}(\theta_{\text{joint}}^*) + O\left(\frac{1}{N}\right), \quad (3.2)$$

the first term the permanent price, the second a transient that shrinks with ramp length. A slow ramp is precisely what the λ -curriculum of this thesis implements.

3.2 The Two-Level Gap: Bowing and Overshoot

Real trajectories deviate from the reference path on two distinct levels, and the step size is what separates them.

Level One (the intended trajectory and its bow): The *intended trajectory* is the path exact gradient descent would follow on the replay objective: gradient flow, the limit of vanishing step size, computed with exact gradients. Even this idealised path bows. The steepest-descent direction of the abruptly assembled joint loss misaligns with the direction toward the joint minimum whenever the loss curves more sharply in some directions than in others, so the path leaves the old task’s basin before returning [15]. The gradient imbalance at the task boundary belongs to this level. At θ_A^* the replay gradient vanishes because the iterate sits at the replay optimum, while the new-task gradient is large [4]; the intended trajectory therefore sets off along the new-task direction, and the imbalance shapes where the bow bends. It is a property of the vector field the descent follows, not of how that field is discretised. This deterministic out-and-back arc forms the structural core of the gap (Figure 1.1, green curve).

Level Two (Overshoot of the followed path): The *followed path* is the trajectory the optimiser actually traces. It deviates from the intended trajectory through two fundamental sources of error that only manifest at a finite step size η : discrete updates cut straight chords across the curving landscape (geometric errors that momentum further accumulates), and sampling noise corrupts the buffer-estimated replay gradient (Section 2.1). A single parameter therefore governs this entire level, acting through two distinct drivers. The *acute driver* is the finite step size meeting the massive, unopposed boundary field: the very first update commits to the full new-task gradient over a displacement of $\eta \|g_{\text{cur}}\|$, causing the iterate to jump along the new-task direction before the growing replay gradient can pull it back. Within a handful of steps, it sits much further out than the intended trajectory ever goes, producing the sharp first-step drop. The

chronic driver is the ongoing sampling noise, which corrupts the restoring gradient at every step, slows the recovery tail, and can trigger sudden drops long after the boundary has been crossed.

We hypothesise that these two levels shape the two dimensions of the stability gap: depth records the overshoot, whether from the acute boundary spike or a late, noise-driven drop, while the hanging tail consists of the Level-One arc combined with the slow, noisy recovery the chronic driver leaves behind. This mapping forms our falsifiable core, formalised as RQ1 in Section 4: interventions on the step, slowing it or denoising it, should mitigate the drop but leave the arc intact; placing the optimiser onto the reference path should instead remove the arc; and only combining both should close the gap completely, while the permanent cost of Section 3.1 remains.

3.3 Two Gates: A Division of Labour

Equation (2.1) shows that replay methods differ mainly in how they weight the replay and current-task gradients. Only the ratio of the two weights sets the update *direction*; their common scale sets the length of the step. In gradient flow the scale can be absorbed into the learning rate exactly, so any such method can be rewritten with the replay weight normalised to one. At a step size greater than zero, however, the absorbed scale *is* the effective learning rate, and Section 3.2 identifies the step size as the acute Level-Two driver; the choice of scale is therefore not harmless. We keep the normalisation but fix the learning rate across all methods and conditions, so every comparison runs at a matched step size. Furthermore, we measure the step size’s own contribution in a dedicated block, the step-size ladder of Section 5, which drives η towards zero at matched flow time and leaves the objective, the buffer, and the update direction untouched [22].

The replay gradient is the only restoring force pulling the iterate back to the reference path, so a protective method must attenuate the new-task push, the source of the bow and overshoot, while leaving the restoring gradient intact. Generalising the current-task weight to an operator G yields a one-sided *gate*:

$$d = G g_{\text{cur}} + g_{\text{rep}}. \quad (3.3)$$

The curriculum of Section 3.1 realises this gate along one axis, *time*: it introduces the new task slowly, so the push is never large enough at once to knock the iterate off the reference path. A complementary axis is *direction*. At any single step only part of the push is harmful, because the past task is flat in most directions and steeply curved in a few, and only motion along the steep directions raises the replay loss. Preconditioning the push by past-task curvature exploits this, passing it at full gain where the past task is flat and shrinking it where the past task is steep, so retention costs little plasticity. This natural-gradient idea

underlies Natural Continual Learning [15] and, in a replay setting, Online Curvature-Aware Replay [27]. Both, however, precondition the *entire* update, so the same metric that brakes the harmful push also brakes the restoring replay gradient, protecting the past task only by slowing the return to it. Asymmetric PER keeps the gate one-sided: it filters the new-task gradient alone and leaves the restoring gradient at full strength. We include the symmetric preconditioner, which damps both, as a control (Appendix C.5).

This thesis explores two specific gating mechanisms to implement this attenuation, alongside a fixed linear ramp that serves as an ablation:

Adaptive λ -curriculum: A scheduling parameter $\lambda(t)$ sets the new-task weight from the ratio of replay to new-task gradient norms, a live damage signal that fires whichever direction harm arrives from. By scaling the loss itself, it deforms the objective into the homotopy of Section 3.1, so the optimiser’s natural descent follows the reference path.

Asymmetric PER: The gate $G = \delta(F_{\text{rep}} + \delta I)^{-1}$ is a filter set by the replay Fisher F_{rep} , giving the update

$$d = \delta(F_{\text{rep}} + \delta I)^{-1} g_{\text{cur}} + g_{\text{rep}}, \quad (3.4)$$

with the damping δ its single knob: as $\delta \rightarrow 0$ the push concentrates in the directions the past task is flat in. The filter is rebuilt each step from the current parameters and past-task curvature.

These distinct mechanisms yield specific hypotheses regarding how each gate navigates the stability-plasticity dilemma, formalised as research questions in Section 4. While both gates aim to steer the intended trajectory back toward the monotone reference path, they tackle the Level Two deviations from fundamentally different angles. Because the curriculum holds back the entire new-task push at the source, it successfully mitigates the acute boundary overshoot, but it does so at a direct plasticity cost. Furthermore, it leaves chronic sampling noise exposed, which can lead to late, noise-driven drops. Consequently, it requires a separate accelerator to repay the plasticity debt (RQ2). Conversely, the asymmetric filter isolates the harmful push by direction, theoretically managing the noise and protecting the past task with a minimal plasticity penalty (RQ3). However, this step-level filtering leaves the method prone to acute overshoot at the start. This theoretical elegance also trades off against computational cost: while the curriculum requires only a cheap gradient-norm ratio per step, asymmetric PER must solve a linear system in the replay Fisher via conjugate gradients (Appendix A.5).

Crucially, we hypothesise that optimiser momentum will expose a failure axis that discriminates between the two gates based on *where* they attenuate

the gradient. Because the curriculum attenuates at the *source* ($\lambda \approx 0$ early in training), the new-task gradient is structurally flattened at the loss level. With no large gradient to accumulate, momentum can safely compose with the gate. Here, momentum solves both of the curriculum’s weaknesses simultaneously: it provides acceleration to repay the plasticity debt, and it acts as a smoothing mechanism to average out the chronic gradient noise, all without re-inflating the acute overshoot. In contrast, asymmetric PER attenuates at the *step*. The raw new-task gradient remains large, and the filter shrinks it step-by-step. Because momentum accumulates past velocities, it ultimately rebuilds the harmful displacement that the per-step filter attempts to remove. For the directional gate, momentum is therefore expected to restore plasticity while actively worsening the acute overshoot, undermining the initial protection and deepening the gap.

This division of labour illustrates that mitigating the stability gap is about understanding the specific mechanisms at play. These two approaches reveal how their specific gating mechanisms interact with the momentum regime.

4 Research Questions

The two-level account and tracking bound of Section 3 make falsifiable predictions about the stability gap and the behaviour of distinct gating mechanisms. To evaluate these predictions, our experiments are organised around four core questions. The first three are evaluated in a controlled setting (rotated MNIST with a small MLP) to probe the gap’s dynamics step-by-step, while the fourth tests the generalisability of these findings under increased complexity.

RQ1: The account. A uniform rescaling of the learning rate is a time reparametrisation of the same gradient flow, so driving $\eta \rightarrow 0$ at matched flow time follows the intended trajectory ever more faithfully without changing it.

Does the stability gap separate under that limit into a deterministic arc that survives it, belonging to the intended trajectory, and an overshoot that exists only at finite step size?

RQ2: The adaptive scalar gate. Taking the limit is itself a repair, but a costly one: a smaller step removes the overshoot and bills for it in training length, the epoch count scaling as $1/\eta$ at matched flow time.

Can an adaptive scalar feedback gate (the adaptive λ -curriculum) fix the trajectory without paying that bill, and how does the answer depend on optimiser momentum and on the plasticity the gate withholds?

RQ3: The directional feedforward gate.

To what extent does a directional feedforward gate (asymmetric PER) reduce the stability gap, and how does accumulated momentum compromise this gating mechanism?

RQ4: Generalisation.

Do the mechanisms underlying the stability gap, and the effectiveness of the proposed gates, generalise to deeper network architectures and more severe task shifts?

5 Methods

This section describes the empirical testbeds (Section 5.1), the metrics that summarise each run (Section 5.1.2), and the experimental blocks (Section 5.2) that answer the research questions of Section 4. All code, experiment configurations, and analysis scripts are available at <https://github.com/Arthurreuss/ValleyWalk>.

5.1 Experimental Setup

5.1.1 Testbeds

Two testbeds of different scale carry the study. A small, fast one (rotated MNIST on a two-layer MLP) runs the dense mechanistic sweeps: the step-size ladder, the curriculum sweep, the asymmetric-PER damping sweep, and the momentum cross. A larger one (domain-shift CIFAR-10 on a CIFAR-adapted ResNet-18) tests whether the findings survive a deeper backbone and a stronger task shift, the regime where the local condition behind the bound of Section 3.1 weakens. Both hold the label space and task structure fixed and shift only the input distribution, and both use two tasks in their main blocks, so there is exactly one transition and one instance of the stability gap; a five-task rot-MNIST sequence and a three-task CIFAR sequence serve as multi-boundary checks.

On rot-MNIST, T_0 presents the digits upright and T_1 rotates them by 90° , each task using the standard 60,000 training and 10,000 test images; the 90° domain shift follows De Lange et al. [4], which orients the pair as $-90^\circ \rightarrow 0^\circ$. On CIFAR-10, T_0 is clean and T_1 applies Gaussian-noise corruption at severity 3 [11]; the ten classes are shared, so only the input distribution shifts. A three-task variant ([none, Gaussian, shot]) and two alternative corruptions appear in the generalisation block. The rot-MNIST MLP is cheap enough to sweep across many conditions and seeds and is where the local condition of Section 3.1 is most plausible; the ResNet-18 is the deeper, more realistic counterpart. rot-MNIST runs online (one epoch per task); the ResNet needs several passes to converge before the transition, so the CIFAR

block uses ten epochs per task. Optimiser, learning rate, and batch size are common across benchmarks; no method retunes the learning rate, so every comparison between methods runs at a matched step size (Section 3.3). The step size is held fixed in that role and manipulated in one block only, the ladder of Section 5.2.1, which holds the objective, the buffer, and the update rule fixed and varies η alone. Full backbone and protocol specifications are in Appendix A.2 (Tables A.1–A.3).

Throughout training every task’s test accuracy is evaluated every 10 steps; at each transition the cadence rises to every step, so the dip is recorded at full resolution (the whole online epoch on rot-MNIST, a 250-step window on CIFAR). A stability-gap tracker snapshots each past task’s pre-switch accuracy and records the resulting (step, accuracy) series. Because the coarse pre-transition cadence leaves a short interval between the last pre-switch evaluation and the first post-update one, the per-step figures hold the last pre-switch level up to the boundary rather than interpolating across that interval; the past task is still training and at its plateau there, so the alternative would draw the drop as beginning before the switch. All reported metrics are computed from the logged samples themselves and are unaffected by this plotting convention. On the vanilla-ER reference runs only, a gradient tracker additionally records the fidelity of the replay-buffer gradient against the exact past-task gradient, at the same cadence; each such step costs a full pass over the past-task training set, which is why it is not run elsewhere, the ladder’s long rungs least of all.

5.1.2 Metrics

Each metric is reported as the seed mean $\pm$ sample standard deviation; complete definitions are in Appendix B. The gap definition of Section 3, the transient excess below the level a task settles at, gives two primary metrics. With a_j^{pre} the pre-switch baseline on past task j and $a_j(t)$ its accuracy at step t , **gap_depth** is the maximum instantaneous excursion $\max_j(a_j^{\text{pre}} - \min_t a_j(t))$, which Section 3.2 predicts to be overshoot-dominated. The second, **gap_area_end**, integrates the shortfall below the settled level: with reference $b_j = \min(a_j^{\text{pre}}, \bar{a}_j)$, where $\bar{a}_j$ is the accuracy the task eventually reaches, it sums $\max(0, b_j - a_j(t))$ over the new task’s training steps and across past tasks. Anchoring to the recovered level implements the split of Section 3.1: a permanent step down to a lower plateau contributes to the standard forgetting metrics and near-zero to the area, while a true dip-and-recover registers in full. One consequence, revisited when reading the tables, is that two methods settling at different levels are measured against different lines, so a better-recovering method can carry the *larger* area; a positive area marks transient disruption, not net cost, and we read it with final accuracy.

The arc of Section 3.2 and the slow part of the noisy recovery live here; sharp noise-driven drops register in depth instead. A third metric, `recovery_steps`, and the gradient-fidelity diagnostics are defined in Appendix B.

One clarification is needed for the ladder of Section 5.2.1, the only block in which η itself varies. `gap_depth` is a maximum over accuracies and is therefore η -invariant by construction. `gap_area_end` is not: it is a trapezoidal integral of the drop curve against the step index, weighted by the spacing between records, with units accuracy-steps. Stretching a rung by $c = 0.1/\eta$ stretches its step axis by the same factor, so an identically shaped excursion reads c times larger. Across the ladder the area is therefore taken in flow time, $A_\tau = \eta A_{\text{steps}}$, the integral of the same curve against $\tau = \sum_t \eta$; within a rung the two differ by a constant, and every block at fixed η reports the step-axis value as before.

To place the gap in the standard frame we report the continual-evaluation metrics of De Lange et al. [4] from the task-boundary accuracy matrix: average final accuracy (ACC), which checks that a gate lowers the gap at preserved accuracy; worst-case retention (min-ACC, WC-ACC); and short-horizon plasticity (WP₁₀₀). FORG and the windowed WF_w/WP_w family are logged for completeness. Conditions are compared with the paired Wilcoxon signed-rank test on seed-paired differences, preferred over the t -test because five paired seeds cannot support a normality assumption on bounded metrics: the gap metrics floor at zero, where the strongest conditions cluster, and accuracy sits near its ceiling, so the seed differences are skewed rather than Gaussian. With five paired seeds the smallest two-sided p -value is 0.0625, reached when all five differences share a sign; we read $p = 0.0625$ as complete directional agreement and let effect sizes separate large results from marginal ones.

5.2 Experiments

Each block maps to one or more research questions and tests the corresponding predictions of Section 3. All conditions use five seeds and the protocol above unless noted. Vanilla ER at the standard operating point (D1: 1k reservoir, $\eta = 0.1$, no schedule and no filter) is the common reference for every rot-MNIST block and, like the blocks it anchors, is run at both momentum settings.

5.2.1 The Step-Size Ladder (RQ1)

RQ1 asks for a limit, and the ladder takes it. A uniform rescaling of η does not change the intended trajectory: forward Euler at step η integrates the same flow $\dot{\theta} = -g(\theta)$ on a finer grid, so a smaller step follows the same path more accurately and reaches the same place having covered the same flow time $\tau = \sum_t \eta$. The ladder is therefore an isolation rather than one

more comparison. Whatever shrinks as η falls is discretisation, Level Two by definition; whatever survives belongs to the path the optimiser was trying to follow all along, which is the arc. The curriculum and every rebalancing of the two gradients deform the vector field itself and so cannot make this separation; the step size is the one knob that leaves the field alone.

Four constraints make the four rungs read as a limit rather than as four unrelated runs. *Matched flow time*: the epoch count scales as $1/\eta$, so a rung at $\eta = 0.1/c$ runs c epochs and every cell covers the same $\tau = 23.5$ over the new task. Without this a small- η cell simply does not finish the transition, and its small gap is under-training rather than a result. *A frozen θ_0^** : every cell warm-starts from one shared task-0 checkpoint (Appendix A.4) rather than training T_0 at its own step size. Training T_0 at the cell's own η would land it in a differently sharp minimum, since less gradient noise means less implicit flattening, and the replay loss accelerates along the trajectory as $\hat{\theta}^\top H_{\text{rep}} \hat{\theta}$, so a sharper past-task Hessian *enlarges* the arc: the confound points in the direction of the hypothesis and has to go. With it gone, θ_0^* and the replay buffer are identical across the four rungs at a given buffer arm, momentum, and seed, and η is the only difference between them. *A matched record grid*: the dense evaluation cadence is also c , so every cell records the same number of samples at the same spacing in flow time. *A cadence counted from the boundary*: inside the dense window the cadence fires on the step index within the task rather than on the global step, because c need not divide the 235-batch epoch and a global-step modulus would otherwise place the first post-switch sample at an arbitrary offset into the task, skipping exactly the steps the spike lives in.

Table 5.1 gives the rungs: $\eta = 0.1$ at one epoch, the standard operating point of every other block; $\eta = 0.1/3 \approx 0.033$ at three; $\eta = 0.01$ at ten; and $\eta = 0.001$ at a hundred. The first three settle the $\mu = 0$ limit; the fourth is there for the momentum leg, which needs a further decade before it converges for the reason given below. Each runs on two buffer arms, the 1k reservoir of the standard protocol and a 60k buffer whose replay gradient is the exact empirical past-task gradient (Appendix A.3), and at momentum 0.0 and 0.9. The exact arm switches the estimator variance off at source, so an excursion that survives both the small step and the exact gradient is left with no Level-Two driver to attribute it to.

The momentum leg carries a second reading. Momentum multiplies the asymptotic step by $1/(1 - \mu)$, so the $\mu = 0.9$ rungs sit at effective steps 1.0, 0.33, 0.1 and 0.01 against the $\mu = 0$ rungs' 0.1, 0.033, 0.01 and 0.001. Two diagonals are therefore exact, ($\eta = 0.01, \mu = 0.9$) against ($\eta = 0.1, \mu = 0$) and ($\eta = 0.001, \mu = 0.9$) against ($\eta = 0.01, \mu = 0$), and they are what the reading rests on. The same arithmetic is why the momentum leg needs the fourth rung

Label	η	Epochs c	Cadence	Steps	τ
S1	0.1	1	1	235	23.5
S2	0.1/3	3	3	705	23.5
S3	0.01	10	10	2350	23.5
S4	0.001	100	100	23500	23.5

Table 5.1: The step-size ladder on two-task rot-MNIST. Cadence is the dense post-switch evaluation interval in steps, counted from the task boundary; it tracks $c = 0.1/\eta$, so all rungs record the same 235 samples over T_1 at the same spacing in flow time $\tau = \sum_t \eta$. Every rung runs on both buffer arms, the 1k reservoir and the 60k exact past-task gradient, and at momentum 0.0 and 0.9: sixteen cells, five seeds, all warm-started from the same frozen θ_0^* per momentum setting and seed.

at all: its three coarsest cells span the effective range in which the $\mu = 0$ leg had not yet converged. A momentum cell that behaves like its matched $\mu = 0$ partner locates momentum’s boundary signature in the effective step; one that does not locates it in the accumulator. Momentum is not a reparametrisation of the flow, however, so the $\mu = 0.9$ leg is flow-matched only within itself: comparisons are drawn within a momentum setting, and the matched-step diagonal is a secondary read.

Because forward Euler’s global error is $O(\eta)$, a residual metric should behave as $A(\eta) = A_0 + k\eta$ over the small- η rungs, and the limit is estimated by fitting that line rather than asserted from the smallest rung; the fit returns A_0 with an error bar, and a departure from linearity at $\eta = 0.1$ is itself the signature of the discrete instability the account holds responsible for the first-step spike. The two levels then predict opposite behaviour for the two metrics. Depth, which records the overshoot, should fall towards zero down the ladder, and fastest on the exact arm. The area in flow time, which carries the arc, should not: it should converge on a positive A_0 that no further reduction of the step size reaches.

5.2.2 The λ -Curriculum Sweep (RQ2, RQ1)

The curriculum sweep tests the scalar feedback gate one ingredient at a time. The linear ramps C1–C3 test the $O(1/N)$ prediction of Equation (3.2). The ladder of Section 5.2.1 sets the bar they have to clear: shrinking every step suppresses the overshoot too, and does so with no method at all, but it buys the reduction with a run stretched by the same factor and with whatever accuracy the shortened steps fail to reach in the budget. The curriculum’s claim is therefore not that it lowers the gap but that it lowers the gap at the standard step size, in the standard number of steps, and on one side of the update only. All conditions run on top of standard ER (1k reservoir, $\eta = 0.1$, the two gradients summed as they come), so the schedule is the only changed variable (Table 5.2). The adaptive

schedule (C4) is the feedback variant of Section 3.3, a clipped exponential moving average of the gradient-norm ratio; it starts near 0 at θ_0^* by stationarity and self-terminates when the gradients rebalance, with no ramp length to choose (mechanics in Appendix A.3). The floor $\lambda_{\min}$ (C5–C7) keeps the schedule off $\lambda \approx 0$ during the first steps and buys back early plasticity. These two are design details of the shipped configuration; the ingredient sweep that settles them is reported in Appendix C.2.

C8 combines the practical configuration (C7) with the exact 60k replay gradient of the ladder’s exact arm, fixing the path *and* switching the estimator variance off at source. Its role is a substitution argument. At momentum 0 the curriculum leaves a residual transient; if that residual is buffer noise, the exact gradient should remove it, and what C7 is missing is then not a second repair of the path but a variance reducer. Only one is affordable at a 1k memory. Momentum averages successive replay estimates inside the velocity buffer and so lowers the variance of the restoring gradient while storing no additional samples, which makes $C7_M$ the cheap substitute for C8’s 60k buffer rather than a fortunate pairing. The comparison also splits momentum’s two roles, since a buffer that leaves no variance to reduce isolates whatever momentum still contributes as new-task acceleration. C8 breaks the limited-memory premise and is diagnostic only; C7 remains the practical configuration.

5.2.3 Momentum Cross (RQ2, RQ3)

The vanilla-ER reference, every cell of the step-size ladder S1–S3, every curriculum condition C1–C8, and the asymmetric-PER sweep legs P1–P2 at the settings noted in Table 5.3 are run at momentum 0.0 and 0.9; momentum-on runs carry an M subscript (e.g. $S1_M$). The cross tests the composability prediction of Section 3.3: whether momentum helps or hurts a method’s gap should depend on where the method attenuates the push. A five-task rot-MNIST sequence (rotations $[0, 30, 60, 90, 120]^\circ$, four transitions, five seeds) repeats the three-way comparison of vanilla ER, the practical curriculum, and asymmetric PER at both momentum settings, as a multi-boundary check (Appendix C.9).

5.2.4 Asymmetric-PER δ Sweep (RQ3)

The damping sweep tests the directional feedforward gate: does the gap shrink as the filter strengthens, at what plasticity cost, and where does the gate fail? Asymmetric PER implements Equation (3.4) [13]; along an eigendirection of replay curvature ρ the gain is $\delta/(\rho + \delta)$, so $\delta \rightarrow \infty$ recovers plain ER and $\delta \rightarrow 0$ concentrates the push in replay-flat directions, with the restoring gradient g_{rep} at full strength throughout (Section 3.3). The push is applied matrix-free by conjugate gradients over Fisher-vector products,

Label	Condition	$\lambda(t)$ schedule	Hyperparameter	Tests
—	Vanilla ER (baseline)	$\lambda \equiv 1$ from step 1 (abrupt switch)	—	reference
C1	Linear $N=50$	$\text{clip}(t/50, 0, 1)$	$N = 50$	RQ2
C2	Linear $N=100$	$\text{clip}(t/100, 0, 1)$	$N = 100$	RQ2
C3	Linear $N=200$	$\text{clip}(t/200, 0, 1)$	$N = 200$	RQ2
C4	Adaptive	$\text{clip}(\text{EMA}_\alpha(r), 0, 1)$	$\alpha = 0.05$	design
C5	Adaptive + $\lambda_{\min}=0.05$	$\max(\lambda_{\min}, \text{clip}(\text{EMA}_\alpha(r), 0, 1))$	$\alpha = 0.05, \lambda_{\min}=0.05$	design
C6	Adaptive + $\lambda_{\min}=0.10$	as C5	$\lambda_{\min}=0.10$	design
C7	Adaptive + $\lambda_{\min}=0.20$	as C5	$\lambda_{\min}=0.20$	design
C8	C7 + full 60k buffer	as C5, exact replay gradient	$\lambda_{\min}=0.20$	RQ2

Table 5.2: λ -curriculum conditions, all applied on top of standard ER. EMA_α is an exponential moving average with smoothing α , and $r := \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ is the gradient-norm ratio. The conditions marked *design* settle the two ingredient choices of the shipped configuration and are reported in Appendix C.2. Each condition is run at momentum 0.0 and 0.9.

Label	Condition	Replay buffer	Damping δ	Purpose
P1	Standard leg	1k reservoir	1.0, 0.3, 0.1, 0.03	practical regime, matched to the curriculum (RQ3)
P2	Full-buffer leg	60k (exact g_{rep})	1.0, 0.3, 0.1, 0.03	variance driver off; residual area is the arc (RQ3, RQ1)
P3	Solver control	60k	0.03	60 CG iterations instead of 25; rules out solve artefacts
P4	Symmetric control	1k reservoir	1.0, 0.3, 0.1, 0.03	preconditions $g_{\text{cur}} + g_{\text{rep}}$; ablates the one-sided gate

Table 5.3: Asymmetric-PER conditions. Both sweep legs run at momentum 0.0; momentum-0.9 counterparts cover $\delta \leq 0.3$ on the standard leg and $\delta \in \{0.3, 0.1\}$ on the full-buffer leg. The two controls run at momentum 0, where the effects they check for arise. The filter metric is always the replay Fisher F_{rep} , estimated on a sampled batch (Appendix A.5).

and the Fisher is re-estimated at every step from the current parameters on a sampled replay batch (Appendix A.5). When momentum is enabled, it acts downstream of the gate: the filtered direction enters the velocity accumulator, and the accumulated velocity is applied unfiltered; preconditioning the velocity instead is a distinct design, discussed in Section 7.5.

Table 5.3 lays out the sweep: $\delta \in \{1.0, 0.3, 0.1, 0.03\}$ crossed with two buffer legs, each at both momentum settings. The *standard* leg (1k reservoir) matches the curriculum’s practical regime, while the *full-buffer* leg makes the replay gradient exact, switching off the variance driver as on the ladder’s exact arm, so the remaining area measures the deterministic arc and the sweep reads out directly whether the gate shrinks the arc itself. Two controls guard the reading: a solver control reruns the strongest filter at 60 CG iterations to separate gate properties from an under-converged solve (Appendix A.5), and a symmetric control preconditions the summed gradient by the same replay Fisher, isolating the effect of leaving the restoring gradient untouched (Appendix C.5).

5.2.5 Generalisation: CIFAR-10 (RQ4)

The CIFAR-10 block tests whether both gates and the momentum signatures carry to a deeper backbone and a stronger task shift. All CIFAR conditions use the ResNet-18 protocol of Table A.3 (ten epochs per

task, buffer 5,000). The headline conditions run at momentum 0.9 (Table 5.4, top), each method crossed with BatchNorm and GroupNorm because the two respond differently at the switch: the input shift disrupts BatchNorm’s running statistics, which must re-adapt during the first steps, while GroupNorm uses no cross-batch statistics, so its transient reflects optimisation dynamics alone. The curriculum runs as the adaptive schedule at $\lambda_{\min} = 0.20$; asymmetric PER runs at $\delta = 0.1$, a strong rot-MNIST setting. Both carry their rot-MNIST configurations without a CIFAR-specific sweep, so they meet the deeper backbone on equal footing. A momentum cross (Table 5.4, bottom) adds momentum-0 runs of all three methods at GroupNorm, repeating the three-way composability test where the transient is free of normalisation-statistics effects; because momentum also drives convergence on this backbone, accuracy is compared within a momentum setting. A further generalisation block pairs the practical curriculum against a paired vanilla-ER control on shifts it was never tuned on (a two-task shot-noise shift, a two-task contrast shift, and a three-task [none, Gaussian, shot] sequence, each at both normalisations over three seeds) to test whether the result depends on the particular corruption or on having a single transition.

Label	Method	Norm	μ
G1	Vanilla ER	BatchNorm	0.9
G2	Adaptive $\lambda_{\min}=0.20$	BatchNorm	0.9
G3	Asym. PER $\delta=0.1$	BatchNorm	0.9
G4	Vanilla ER	GroupNorm	0.9
G5	Adaptive $\lambda_{\min}=0.20$	GroupNorm	0.9
G6	Asym. PER $\delta=0.1$	GroupNorm	0.9
G7	Vanilla ER	GroupNorm	0.0
G8	Adaptive $\lambda_{\min}=0.20$	GroupNorm	0.0
G9	Asym. PER $\delta=0.1$	GroupNorm	0.0

Table 5.4: CIFAR-10 conditions (two-task Gaussian-noise shift, buffer 5,000, five seeds). Top: the headline block at momentum 0.9. Bottom: the momentum cross, pairing momentum-0 runs of all three methods against their momentum-0.9 counterparts at GroupNorm. The generalisation block (novel corruptions and the three-task sequence, three seeds) is described in the text and reported in Appendix C.7.

6 Results

The results follow the research questions: the two-level account (Section 6.1, RQ1), the scalar feedback gate (Section 6.2, RQ2), the directional feedforward gate (Section 6.3, RQ3), a synthesis of the two momentum crosses (Section 6.4), and the CIFAR-10 generalisation of both gates (Section 6.5, RQ4).

Reading the tables. Depth is in percentage points (pp): a depth of 19.5 means the worst past-task evaluation falls 19.5 points below its pre-switch baseline. The end-referenced area `gap_area_end` integrates the excursion below the level each task settles at; its conventions, and the caveat that cross-condition areas must be read together with final accuracy, are stated once in Section 5.1.2. Values are mean $\pm$ standard deviation over five seeds (three for the CIFAR generalisation block); comparisons follow the Wilcoxon convention of Section 5.1.2, quoting a p -value only where seeds disagree. The full metric set for every condition is in Appendix C.

6.1 The Two Levels, Separated (RQ1)

RQ1 asks whether the transient separates into a deterministic arc of the intended trajectory and overshoot of whatever path is actually followed. The ladder of Section 5.2.1 answers it as a limit rather than as a comparison: with the objective, the update rule, θ_0^* and the replay buffer bit-identical across rungs and only η varying at matched flow time, the four rungs are four discretisations of one and the same intended trajectory, and whatever survives $\eta \rightarrow 0$ is that trajectory. Depth is η -invariant and quoted in pp throughout; the area is the flow-time integral $A_\tau = \eta A_{\text{steps}}$, the only form comparable across rungs (Section 5.1.2).

The ladder converges, and not on zero. At momentum 0 the three finest rungs are all but the same run. On the exact arm depth falls $22.3 \rightarrow 4.2 \rightarrow 3.9 \rightarrow 3.8$ pp for $\eta = 0.1 \rightarrow 0.1/3 \rightarrow 0.01 \rightarrow 0.001$, with the flow-time area at $0.590 \rightarrow 0.360 \rightarrow 0.346 \rightarrow 0.341$ and ACC at $0.895 \rightarrow 0.897 \rightarrow 0.897 \rightarrow 0.898$; on the sampled arm, $22.9 \rightarrow 5.6 \rightarrow 4.9 \rightarrow 4.7$ pp, area $0.577 \rightarrow 0.293 \rightarrow 0.275 \rightarrow 0.270$, ACC $0.874 \rightarrow 0.876 \rightarrow 0.877 \rightarrow 0.877$ (all sixteen cells with their full metric set in Appendix C.1). The last step down each ladder, a further tenfold cut from $\eta = 0.01$ to $\eta = 0.001$, moves depth by 0.1 and 0.2 pp against the 18.1 and 17.3 pp the first step removes, and the trajectories themselves coincide: the largest separation between the $\eta = 0.1/3$ and $\eta = 0.01$ curves anywhere on the common flow-time grid is 0.007 (exact) and 0.006 (sampled) in accuracy, against 0.201 and 0.219 for $\eta = 0.1$, both attained at $\tau \approx 0.2$, inside the coarse rung’s first two steps (Figure 6.1). These residual differences are small but not noise: three of the four contrasts between those two rungs are unanimous in sign across seeds ($p = 0.0625$; the exception is the sampled area, four of five, $p = 0.125$), which is what an $O(\eta)$ remainder should look like. Extrapolating $A(\eta) = A_0 + k\eta$ through the fine rungs alone puts the limit at 3.8 pp of depth and 0.34 of area on the exact arm, 4.7 pp and 0.27 on the sampled one. That positive limit is RQ1’s answer: an excursion that survives both the exact gradient and the vanishing step has no Level-Two driver left to attribute it to, and it is the arc of Section 3.2 measured rather than inferred. The all-rung least-squares fit is not the estimate to read here: over a $100\times$ lever arm the line is pinned by the coarse rung, whose excess inflates the slope to $k \approx 2.0$, pulls the depth intercept down to 1.5 ± 1.1 pp, and leaves a largest residual of 3.8 pp, the size of the very quantity being estimated. That is the linear model reporting that one of its four points does not belong to its regime.

The coarse rung sits above the line. How far it sits above is the sharpest single number the block produces: 17.2 pp of depth and 0.19 of area on the exact arm, 15.6 pp and 0.24 on the sampled one. This excess is by construction what the $O(\eta)$ model cannot account for, and it is the discrete instability itself, so between two thirds and three quarters of the depth measured at the standard operating point is not a property of the trajectory the optimiser is following. It is not estimator noise either: the departure on the exact arm, where the replay gradient carries no sampling error at all, is the larger of the two in depth. This is also where the ladder settles what the boundary imbalance is. The imbalance is identical at every rung, the same near-zero replay gradient meeting the same large new-task gradient at the same θ_0^* , since a rescaling of η leaves the vector field untouched; yet the spike it produces is gone by $\eta = 0.1/3$ and the

Driving the step to zero: what vanishes is discretisation, what remains is the arc

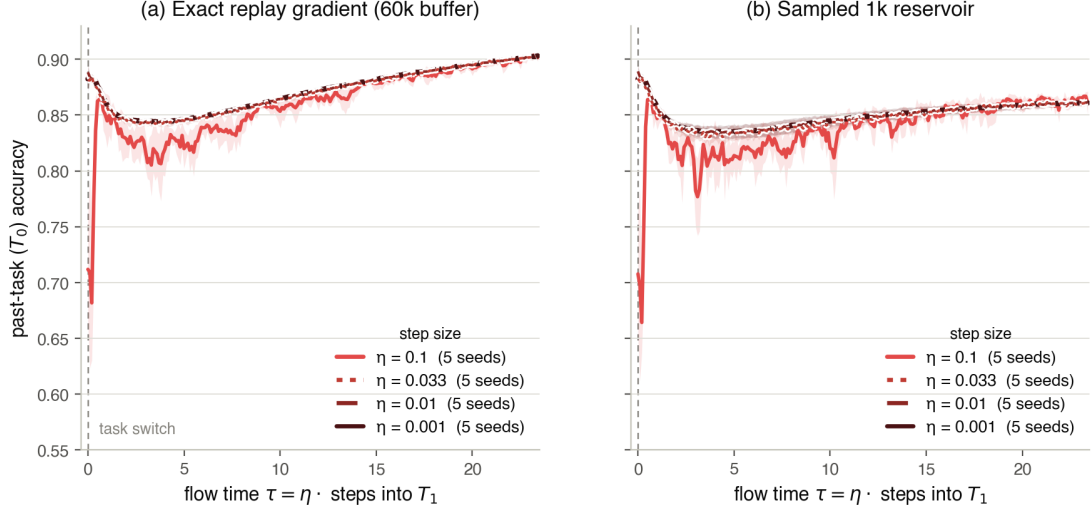


Figure 6.1: The step-size ladder at momentum 0: past-task (T_0) accuracy against flow time $\tau = \sum_t \eta$, seed mean with $\pm \text{std}$ bands, $\tau=0$ the task switch. Every rung covers the same $\tau = 23.5$ from the same frozen θ_0^* and the same buffer, so within a panel η is the only difference. (a) exact 60k replay gradient, (b) the 1k reservoir of the standard protocol. What the reader should see is a limit that is reached and is not zero. The standard operating point (light, $\eta = 0.1$) leaves the other three within its first two steps, drops ≈ 20 pp, and spends the rest of the task jittering back; the three finer rungs (dark: $\eta = 0.1/3$ and $\eta = 0.01$ dashed, $\eta = 0.001$ solid) have no boundary spike at all, and overlap so closely that linestyle rather than colour is what separates them. What they do have is a smooth dip-and-recover bend that neither the smaller step nor the exact gradient removes: the arc.

excursion that remains is smooth. The imbalance is Level-One terrain, and the acute driver of the drop is the step size that meets it, exactly the division of labour Section 3.2 assigns them. The measured limit does qualify the metric mapping of that section, however: depth is overshoot-dominated at the operating point, but it is not overshoot alone, because the arc still registers 3.8 pp of depth once the overshoot is gone.

Accuracy is not what the limit costs. Nothing on the ladder is traded away for the smaller gap. ACC is flat to slightly rising down both arms ($0.895 \rightarrow 0.898$ exact, $0.874 \rightarrow 0.877$ sampled), and min-ACC, the worst single evaluation of the run, improves sharply, $0.658 \rightarrow 0.843$ and $0.653 \rightarrow 0.834$. (WP_{100} falls, $0.453 \rightarrow 0.376$ and $0.454 \rightarrow 0.369$, but it is a fixed hundred-*step* horizon and therefore spans a hundredth of the flow time at $\eta = 0.001$; the decline is a units artefact of the reparametrisation and is not read as a plasticity cost.) What the limit costs is compute. Matched flow time makes the epoch count scale as $1/\eta$, so the $\eta = 0.001$ rung removes 18.5 pp of depth at a hundred times the training of the operating point, and it removes only the overshoot. This is the bar the gates of RQ2 and RQ3 have to clear: not lowering the gap, which the step size already does with no method at all, but lowering it at the standard step size and in the standard number of steps.

Under momentum the ladder converges too, on an arc of its own. Momentum multiplies the asymptotic step by $1/(1 - \mu)$, so the four rungs sit at effective steps 1.0, 0.33, 0.1 and 0.01, and only the last of these reaches the range in which the $\mu = 0$ leg had settled. It does reach it. Depth falls $14.1 \rightarrow 9.9 \rightarrow 5.0 \rightarrow 2.7$ pp on the exact arm (area $0.301 \rightarrow 0.122 \rightarrow 0.061 \rightarrow 0.040$, ACC $0.965 \rightarrow 0.973$, min-ACC $0.814 \rightarrow 0.928$) and $15.4 \rightarrow 10.1 \rightarrow 5.4 \rightarrow 4.6$ pp on the sampled one (area $0.105 \rightarrow 0.021 \rightarrow 0.0034 \rightarrow 0.0006$, ACC $0.929 \rightarrow 0.942$, min-ACC $0.802 \rightarrow 0.910$). The fine-rung extrapolation puts the limit at 2.6 pp of depth on the exact arm and 4.1 pp on the sampled one, only 0.1 and 0.5 pp below the finest rung actually run, so this leg has an arc and the bottom of the ladder is close to it. What remains true of the earlier reading is narrower than it was: the coarse rung's departure is still *negative* (-10.3 pp exact, -6.2 pp sampled), lying below the line through the fine rungs rather than above it as at $\mu = 0$, so the all-rung fit is still not the estimate to read. The $O(\eta)$ model breaks down at the coarse end of both legs; under momentum it breaks down by saturating rather than by exploding.

Momentum is not simply a longer step. The grid now provides two matched-effective-step diagonals, and together they say what one alone could not. At effective step 0.1, ($\eta = 0.01, \mu = 0.9$) against

($\eta = 0.1, \mu = 0$) separates widely, 5.0 pp / 0.061 / 0.972 against 22.3 pp / 0.590 / 0.895 on the exact arm; at effective step 0.01, ($\eta = 0.001, \mu = 0.9$) against ($\eta = 0.01, \mu = 0$) reads 2.7 pp / 0.040 / 0.973 against 3.9 pp / 0.346 / 0.897. The depth separation collapses from 17.3 to 1.2 pp as the matched step shrinks, which is what two legs approaching their own $\eta \rightarrow 0$ limits must do; the area separation does not, staying eight-fold at the finest matched pair. What differs between the legs is therefore the shape of the residual excursion and not only its scale. This remains the secondary read Section 5.2.1 calls it: the legs warm-start from anchors trained at their own momentum and settle some seven accuracy points apart, so comparing their levels compares two operating points.

Momentum’s arc is depth without tail. The within-leg contrast is both the safer comparison and the sharper one. On the sampled arm the momentum leg’s flow-time area falls by more than two orders of magnitude across the ladder, $0.105 \rightarrow 0.0006$, and extrapolates to zero (-0.001), while its depth converges on 4.1 pp. In the limit the transient that survives under momentum is a dip that recovers almost at once: it keeps the instantaneous drop and loses the hanging tail. The $\mu = 0$ arc keeps both, 4.7 pp of depth and 0.27 of area that no further shortening of the step removes. Of the two-part measure of Section 5.1.2, then, momentum leaves essentially only the depth term standing.

The coarse rung reproduces the operating point in distribution. $\eta = 0.1$ on the sampled arm is the standard operating point, but it is not the archived vanilla-ER reference run (D1, Appendix C.2) and is not paired with it: the ladder skips T_0 training so that every cell can share one θ_0^* , which leaves the two protocols on different RNG streams and hence with different buffer contents seed for seed. Their distributions agree, 22.9 ± 4.2 against 19.5 ± 4.1 pp of depth and 5.77 ± 0.80 against 6.27 ± 0.47 of step-axis area, one-standard-deviation intervals overlapping on both. That is a check that warm-starting has not moved the operating point, and nothing stronger than that.

What the gates inherit. RQ1 therefore hands on a measured quantity rather than a hypothesis. At momentum 0 the arc is 3.8 pp of depth and 0.34 of flow-time area at ACC 0.898 on the exact arm (4.7 pp / 0.27 / 0.877 sampled): a residual excursion that neither the exact gradient nor any further shortening of the step reaches, because it is not a discretisation error but the path the optimiser was following all along. Removing it requires changing the path, which is what the two gates attempt, and at the standard step size, where the ladder’s own remedy is not available.

6.2 The Scalar Feedback Gate (RQ2)

Table 6.1 reports the curriculum sweep at both momentum settings against vanilla ER. The momentum-off leg tests the schedule in isolation; the momentum-on leg is the practically relevant regime.

The schedule works as predicted, at a momentum-0 price. At momentum 0, mean gap depth falls monotonically with the ramp length, $17.5 \rightarrow 16.2 \rightarrow 14.3$ pp at $N = 50, 100, 200$, as the $O(1/N)$ bound of Equation (3.2) predicts, though successive lengths lie within the seed spread and the reduction is bought with accuracy ($0.851 \rightarrow 0.844 \rightarrow 0.828$) and a stretched tail (per-condition metrics in Appendix C.2). The depth that remains, still in the mid-teens, is the untouched chronic driver: the schedule times the push but leaves the finite-buffer replay estimate exactly as noisy as vanilla’s, so the worst evaluation keeps arriving as a late noise-driven drop, a floor no ramp length can pass and only the exact-gradient diagnostic removes (C8, Table C.3). The two remaining ingredients behave as designed but earn their keep only under momentum. At momentum 0 the adaptive signal removes the knob N and changes nothing else, 12.8 pp / 2.2 / 0.833 against the longest linear ramp’s 14.3 pp / 1.8 / 0.828, with seeds split on the depth difference ($p = 0.44$); that null is specific to this momentum setting, and the momentum leg below separates the same pair unanimously in the adaptive schedule’s favour. The $\lambda_{\min}$ floor is likewise inert here, 13.9 pp / 2.7 / 0.837 at $\lambda_{\min}=0.20$ against 12.8 pp / 2.2 / 0.833 without it, because early T_1 progress at momentum 0 is set by the large boundary gradient the small floor cannot redirect. The momentum-0 curves of Figure 6.2 (dashed) show the underlying trade: a shallower past-task dip than vanilla (a), paid for in slower early T_1 learning (b).

The bar is the ladder’s, and at momentum 0 the gate does not clear it. Lowering the gap is not by itself the gate’s claim, because the step size lowers it with no method at all; the claim has to be that the gate lowers it at the standard step size and in the standard number of steps (Section 6.1). Nor is there an accuracy handicap to discount from the ladder’s numbers: ACC is flat to rising down both arms and min-ACC improves sharply, so what the ladder charges is compute, the epoch count scaling as $1/\eta$ at matched flow time. Read against that bar on the matched 1k arm at momentum 0, the curriculum loses on everything but the budget. C7 leaves 13.9 pp of depth at ACC 0.837 and min-ACC 0.743 in one epoch at $\eta = 0.1$; the $\eta = 0.01$ rung leaves 4.9 pp at ACC 0.877 and min-ACC 0.832 in ten. Measured against its own protocol’s reference the schedule removes 5.6 pp of vanilla ER’s 19.5 and spends 3.6 pp of ACC doing it ($0.873 \rightarrow 0.837$), where the tenfold shorter step re-

Condition	Momentum 0.0			Momentum 0.9		
	gap_depth (pp)	gap_area_end	ACC	gap_depth (pp)	gap_area_end	ACC
Vanilla ER (baseline)	19.5 ± 4.1	6.3 ± 0.5	0.873 ± 0.005	15.9 ± 2.6	1.0 ± 0.2	0.928 ± 0.003
C3: Linear $N=200$	14.3 ± 4.6	1.8 ± 0.6	0.828 ± 0.025	6.2 ± 0.6	0.1 ± 0.0	0.925 ± 0.007
C4: Adaptive	12.8 ± 2.9	2.2 ± 0.5	0.833 ± 0.025	2.5 ± 0.5	0.3 ± 0.1	0.917 ± 0.001
C7: Adaptive $\lambda_{\min}=0.20$	13.9 ± 3.4	2.7 ± 0.6	0.837 ± 0.031	3.7 ± 0.4	0.0 ± 0.0	0.931 ± 0.002

Table 6.1: λ -curriculum key conditions on rot-MNIST against vanilla ER, at momentum 0.0 and 0.9; five seeds: the longest linear ramp (C3), the pure adaptive schedule (C4), and the practical configuration (C7). The scalar feedback gate reaches a few points of depth at matched accuracy only in the momentum-on regime it is intended for. The full sweep (C1–C7), with vanilla ER’s full metric row as the reference, is in Appendix C.2.

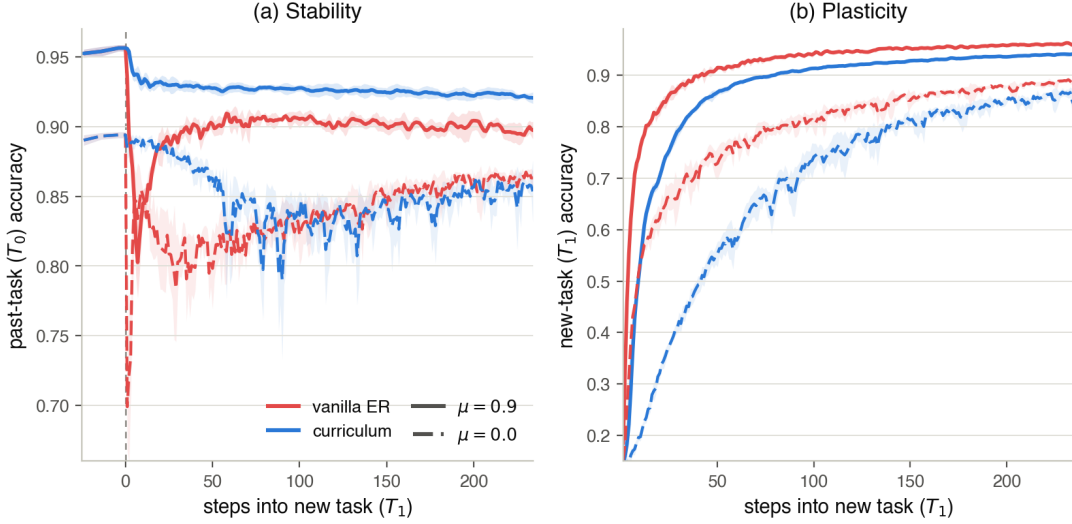


Figure 6.2: The scalar feedback gate and momentum’s two roles: per-step accuracy across the rot-MNIST switch, seed mean with $\pm$ std bands, for vanilla ER (red) and the practical adaptive curriculum $\lambda_{\min}=0.20$ (blue), each at momentum 0.9 (solid) and 0.0 (dashed). (a) Stability (T_0): only curriculum-with-momentum (solid blue) holds the past task near its plateau through the switch; momentum barely dents vanilla’s spike, and both methods open a deep, jittery gap without it (dashed). (b) Plasticity (T_1): at momentum 0 the curriculum slows early new-task learning (dashed blue), the plasticity debt that momentum pays back (solid blue).

moves 18.0 pp of the ladder’s distributionally matched 22.9 and gains 0.3 pp of ACC ($0.874 \rightarrow 0.877$). The flow-time areas do land together, 0.27 for C7 against the $\eta = 0.01$ rung’s 0.275 (the curriculum’s step-axis 2.7 converted at $A_\tau = \eta A_{\text{steps}}$), but the two are referenced to plateaus four points apart and a lower plateau is a lower line to integrate against, so this is not a tie (Section 5.1.2). The momentum-0 case for the scalar gate is therefore a case about cost and not about depth: it buys a fraction of what a shorter step buys, and buys it inside the budget a shorter step cannot keep.

Momentum pays the plasticity debt, and closes the gap. The momentum-on leg is the scalar gate’s intended regime, where momentum acts through two roles at once and where both design details settle. Under momentum the adaptive signal does not merely remove the knob N , it outperforms the best setting of the knob it removes: 2.5 pp / 0.3 / 0.917 against the

tuned $N=200$ ramp’s 6.2 pp / 0.1 / 0.925, shallower in all five seeds by ≈ 3.7 pp, paid for in accuracy and a slightly heavier tail. The $\lambda_{\min}=0.20$ floor buys both back at a quantified price: 3.7 pp / 0.03 / 0.931, unanimously 1.2 pp deeper than the unfloored schedule and unanimously 1.3 pp more accurate, with the tail an order of magnitude lighter. The two ingredients compose into a configuration that clears the tuned linear ramp on all three quantities, depth in five seeds of five and accuracy in four, and that cuts three quarters of vanilla ER’s depth (15.9 pp / 1.0 / 0.928) while returning accuracy to vanilla’s level. The depth collapse is variance reduction, momentum averaging the noisy replay gradient across accumulated velocity once the schedule has tempered the boundary push at source; the accuracy restoration is the separate role, acceleration, repaying the plasticity the schedule borrows.

C8: the pairing is forced, not found. Momentum is not one partner among several that happened

to suit the schedule, and C8 is the step that shows why. Crossing the practical curriculum with the exact 60k replay gradient at momentum 0 collapses the residual the schedule leaves, $13.9 \rightarrow 0.2$ pp of depth and $2.7 \rightarrow 0.05$ of area at ACC $0.837 \rightarrow 0.818$ (Table C.3, Appendix C.2). What C7 still carries at momentum 0 is therefore estimator variance and not a second deformation of the path, so what it needs is a variance reducer rather than a second repair of the trajectory. At a 1k memory exactly one is affordable: a 60k buffer is the negation of the premise, while momentum averages successive replay estimates inside the velocity buffer and stores nothing to do it. C7_M is in that sense the cheap substitute for C8’s buffer, at 1/60th of its memory, and the substitution runs the other way too: with the buffer exact there is no variance left for momentum to reduce, and what it still contributes shows up as accuracy alone, C8_M reaching 0.955 against C8’s 0.818 at 2.0 against 0.2 pp of depth. The two roles are separable, and the schedule needs one of each.

C7_M (3.7 pp / 0.03 / 0.931, min-ACC 0.919) is accordingly the configuration this thesis recommends and the one carried to CIFAR: no other limited-memory condition in the rot-MNIST study reaches a shallower gap without giving up accuracy. It is also where the gate clears the ladder’s bar in its own regime. On the same 1k arm at momentum 0.9 the $\eta = 0.01$ rung reaches 5.4 pp at ACC 0.940 and min-ACC 0.901 after ten epochs, so one epoch of the schedule is 1.7 pp shallower and 1.8 pp better in worst-case retention than ten epochs of a tenfold shorter step, for 0.9 pp of ACC (the two protocols’ $\eta = 0.1$ cells agree here as at momentum 0, 15.4 ± 2.2 against 15.9 ± 2.6 pp at ACC 0.929 against 0.928). That rung is a rung and not a limit, but the ladder’s fourth rung now supplies the limit as well, and the comparison strengthens rather than softens. At $\eta = 0.001$, a hundred epochs, the same arm reaches 4.6 pp at ACC 0.942 and min-ACC 0.910, and the fine-rung extrapolation puts the $\eta \rightarrow 0$ depth of this leg at 4.1 pp (Section 6.1). One epoch of the schedule is thus 0.9 pp shallower than a hundred epochs of a hundredfold shorter step, and its 3.7 ± 0.4 pp sits at or below what shortening the step can ultimately reach here, for 1.1 pp of ACC. The claim is not that the schedule beats the limit, which the seed spread does not support, but that it matches it at one hundredth of the training: what the step size buys at this momentum, it buys at a cost the schedule does not pay.

For completeness the momentum cross also confirms the vanilla-ER null: momentum shifts vanilla depth only from 19.5 to 15.9 pp, smaller than the seed spread with seeds split on its sign ($p = 0.44$), because the deterministic unopposed first step carries no accumulated velocity to average; it does cut vanilla’s tail (area $6.3 \rightarrow 1.0$, all seeds). Momentum smooths vanilla’s recovery, but it leaves the depth, the gap it-

self, intact.

6.3 The Directional Feedforward Gate (RQ3)

RQ3 asks whether asymmetric PER shrinks the gap as its filter strengthens ($\delta \rightarrow 0$), at what plasticity cost, and where a feedforward gate fails. Table 6.2 reports the δ sweep on both buffer legs at momentum 0; the full metric set at both momentum settings is in Appendix C.4. The standard 1k leg is the gate’s practical configuration; the full-buffer leg plays the same diagnostic role as C8, switching the estimator variance off at source so that what remains of the excursion is the arc together with whatever the standard step still overshoots of it, and it does not enter the head-to-head comparisons. Both legs run at $\eta = 0.1$, so their areas are on the step axis; the ladder’s flow-time arc converts to the same axis as $A_{\text{steps}} = A_{\tau}/\eta$.

Monotone, at no measured plasticity cost. On the standard buffer, depth falls from 9.0 to 4.8 pp as δ goes from 1.0 to 0.03 while ACC *rises* from 0.869 to 0.880 and WP₁₀₀ from 0.733 to 0.741; every one of these trends holds across all five seeds (Figure 6.3(a)). To be precise about what is claimed: the filter does slow the new task, but selectively, only along the directions the past-task curvature flags, never by scaling the whole loss down as the scalar gate does. The new-task learning curves show the honest version of this claim (Figure 6.4): at the strongest filter the T_1 curve is visibly slower over roughly the first fifty steps, the expected signature of the selective slowdown, and indistinguishable from vanilla ER thereafter; by the WP₁₀₀ horizon no cost remains, and the roughly 4 pp of accuracy the scalar gate pays at momentum 0 never materialises here. This is the scalar gate’s weakness inverted, and it is what makes the two gates complements. The full-buffer leg turns the filter on the arc: with the variance driver off, the area shrinks with the filter, $2.3 \rightarrow 1.0 \rightarrow 0.4$ for $\delta = 1.0 \rightarrow 0.3 \rightarrow 0.1$ (the $1.0 \rightarrow 0.3$ drop holds across all seeds) at ACC $0.898 \rightarrow 0.900 \rightarrow 0.901$, so the directional gate bends the followed path back toward the reference, as the curriculum does by another route. The contrast that makes the mechanism point is with the ladder. On the same exact gradient, shortening the step converges on 0.34 of flow-time area, 3.4 on the step axis these runs are measured on, and going below that is what no smaller step achieves; the ungated exact-gradient run at this step size reads 5.9 there, and the gate at $\delta=0.1$ takes it to 0.4 without touching the step size at all. The gate reaches the arc itself, which no repair of the step does. This is a mechanism reading rather than an operating point, since the full buffer that isolates the arc defeats the limited-memory premise. Leaving the restoring gradient g_{rep} at full strength is what makes this a net removal of damage rather than a

δ	Standard (1k) buffer				Full (60k) buffer			
	depth (pp)	area	ACC	WP ₁₀₀	depth (pp)	area	ACC	min-ACC
no gate	19.5 ± 4.1	6.3	0.873	0.736	$22.3 \pm 5.0^\dagger$	5.9	0.895	0.658
1.0	9.0 ± 0.6	3.3	0.869	0.733	4.0 ± 1.0	2.3	0.898	0.841
0.3	7.4 ± 1.1	2.3	0.873	0.736	1.5 ± 1.1	1.0	0.900	0.867
0.1	5.9 ± 1.5	1.6	0.876	0.739	2.2 ± 1.6	0.4	0.901	0.860
0.03	4.8 ± 1.9	1.0	0.880	0.741	9.5 ± 8.2	0.4	0.903	0.787

Table 6.2: Asymmetric-PER δ sweep on rot-MNIST at momentum 0, five seeds, against the ungated reference. Depth and area fall as $\delta \rightarrow 0$ while ACC and WP₁₀₀ rise: the directional gate protects the past task at no measured plasticity cost. On the full buffer, the diagnostic leg on which the residual excursion is the arc and the step’s overshoot of it, the area falls to 0.4 at $\delta=0.1$, below the 3.4 the ladder’s arc alone reads on this step axis ($A_\tau = 0.34$, Section 6.1). The far cell is the exception: at $\delta=0.03$ full buffer the mean depth jumps to 9.5 pp with a large std, the cliffs of the text. Areas are on the step axis ($\eta = 0.1$). [†]The full-buffer reference is the ungated exact-gradient rung of the ladder at the same step size (S1, exact arm, $\mu=0$; Appendix C.1); it warm-starts from a frozen θ_0^* and so is not seed-paired with the sweep.

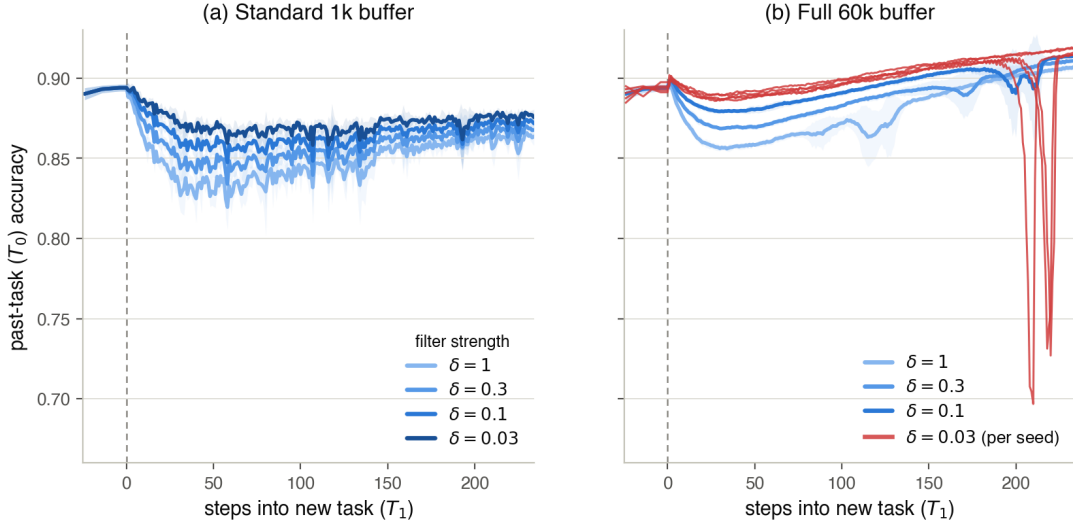


Figure 6.3: The directional feedforward gate: per-step past-task (T_0) accuracy across the rot-MNIST switch at momentum 0, sweeping the damping δ (weaker filter light $\rightarrow$ stronger filter dark). (a) Standard 1k buffer: the boundary dip shrinks monotonically as $\delta \rightarrow 0$, and accuracy rises with it (Table 6.2), so the protection is bought at no plasticity cost. (b) Full 60k buffer (variance driver off, so the residual excursion is the deterministic arc together with the standard step’s overshoot of it): the excursion shrinks with the filter for $\delta \geq 0.1$, but at the strongest setting $\delta=0.03$ the gate fails late and abruptly: shown per seed (critical red), 3 of 5 seeds hold flat for ~ 200 steps and then slide off a late cliff, a failure mode a feedback signal does not share.

reshuffling of it: a symmetric variant that preconditions the summed update strangles the restoring gradient harder than the harmful push, so its forgetting *climbs* as the filter strengthens (Appendix C.5, Table C.6).

The cliffs: an unexpected late failure. The one place the sweep breaks monotonicity is the strongest full-buffer setting, $\delta = 0.03$, where mean depth jumps to 9.5 pp with a std of 8.2: in 3 of 5 seeds the past-task curve holds flat for roughly two hundred steps and then slides off a late cliff, the red per-seed traces in Figure 6.3(b). A solver control rules out the most straightforward explanation: rerunning the setting with the conjugate-gradient solver at 60 itera-

tions instead of 25 reproduces the cliffs at the same seeds and steps (11.5 pp mean depth), so they are not an artefact of an under-converged solve. The cliffs came as a surprise: the account made no prediction of a late, abrupt failure, and nothing in the rest of the sweep hints at one. They are also confined to the exact-gradient diagnostic: at the same δ on the standard 1k buffer no seed shows one (depth 4.8 ± 1.9 pp, the monotone end of the sweep). The failure is tied to the exact replay gradient: the sampling jitter of a practical buffer appears to protect against it, the same noise that elsewhere drives the overshoot here keeping the iterate from settling into the escape direction. Why the cliffs happen we can only partly say. A reading consistent with the data is a blind spot of the

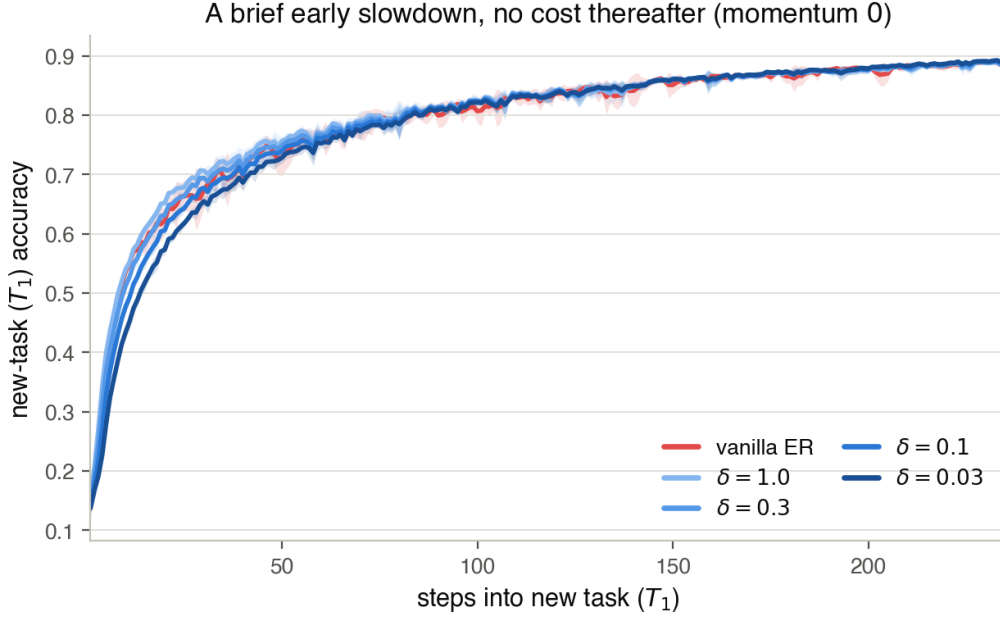


Figure 6.4: New-task (T_1) accuracy across the rot-MNIST switch at momentum 0 for the standard-buffer δ sweep against vanilla ER (red); seed mean with $\pm$ std bands. The strongest filter ($\delta=0.03$, darkest) lags briefly over the first ≈ 50 steps, the selective slowdown along past-task-sharp directions, and matches vanilla from then on; the plasticity metrics at horizon 100 and beyond record no cost (Table 6.2).

gate’s model of the past task: the filter passes new-task pressure through directions its sampled Fisher rates as flat, and in these seeds that trust appears to be misplaced further out along the trajectory. We offer this as a hypothesis, not an established mechanism. What the control does establish is that the strongest filter can fail late and abruptly when nothing perturbs it. Because the failure is confined to the diagnostic leg, it is unlikely in a practical configuration, but it exposes a real weakness of feedforward control.

Momentum re-inflates the per-step filter. Momentum undoes the directional gate’s protection, the opposite of what it does for the scalar gate, and it does so on both legs: on the practical 1k buffer depth climbs from 4.8 to 8.5 pp at $\delta=0.03$ (Figure 6.5), and on the full-buffer diagnostic from 2.2 to 7.6 pp at $\delta=0.1$ and from 1.5 to 9.1 pp at $\delta=0.3$ (both across all five seeds). Momentum accumulates the already-filtered direction, but the filter attenuates the step, not the source: the new-task loss, and with it the per-step gradient, keeps its full size, and because motion in the braked directions is slow, the shrunk push persists there step after step. A persistent push entering the accumulator is amplified toward $1/(1-\mu)$ times its per-step size, so velocity rebuilds displacement in the very directions the filter shrinks one step at a time. This re-inflation is what makes the head-to-head comparison of the next subsection regime-dependent.

6.4 Reading the Momentum Crosses Together

Read side by side in both gates’ practical regime, the 1k reservoir, the curriculum’s and the directional gate’s momentum crosses show that the two gates do not sit on a fixed stability–plasticity frontier a practitioner trades along: which gate leaves the smaller gap is decided by the momentum setting, the same setting that decides final accuracy. Figure 6.5 shows it directly, the same three methods at the same buffer with only the momentum setting changed between the panels.

At momentum 0 the directional gate is better on both axes: 4.8 pp of depth at 0.880 accuracy against the curriculum’s 13.9 pp at 0.837, with no measured plasticity debt to leave unpaid, and its one remaining cost is the per-step conjugate-gradient solve against the scalar gate’s single gradient-norm ratio (Table C.10). Momentum, standard on realistic backbones and decisive for final accuracy at these budgets, reverses the ranking through the re-inflation mechanism above: the directional gate climbs to 8.5 pp at the matched buffer while the curriculum, attenuating at source, reaches 3.7 pp at vanilla accuracy, and with the buffers matched the directional gate has no accuracy advantage left to offset its re-inflated gap (0.929 vs. 0.931). The choice between the gates is therefore not a point on a frontier but a consequence of whether momentum is on: off, take the directional gate; on, take the scalar gate. A five-task rot-MNIST sequence reproduces this picture across four consecutive bound-

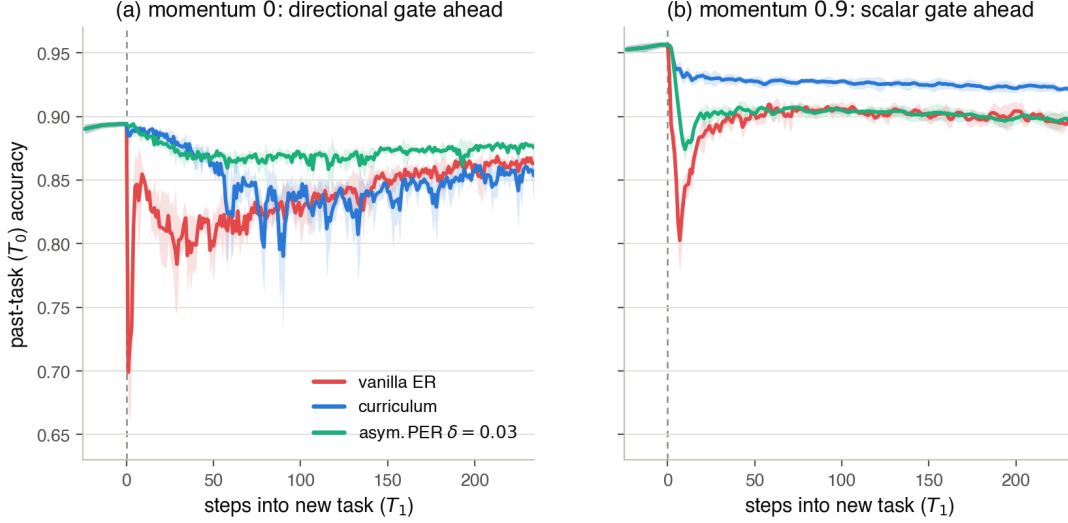


Figure 6.5: The momentum synthesis at the matched 1k buffer: per-step past-task (T_0) accuracy across the rot-MNIST switch for vanilla ER (red), the practical curriculum C7 (blue), and asymmetric PER at its strongest standard-buffer filter $\delta=0.03$ (aqua); seed mean with $\pm$ std bands, $x=0$ the task switch. (a) Momentum 0: the directional gate holds T_0 highest (4.8 pp / ACC 0.880) while the curriculum opens a deep, noisy dip (13.9 pp / 0.837). (b) Momentum 0.9: the ordering swaps: the curriculum holds T_0 near its plateau (3.7 pp / 0.931) while accumulated velocity re-inflates the directional gate’s per-step filter (8.5 pp / 0.929). The pre-switch level is higher in (b) because momentum converges T_0 higher before the switch.

aries: at momentum 0 the directional gate keeps its edge, under momentum the curriculum keeps the least cumulative forgetting while the re-inflated directional gate falls back to vanilla ER’s level (Appendix C.9, Figure C.2).

6.5 Generalisation to CIFAR-10 (RQ4)

The CIFAR-10 block tests whether both gates carry to a ResNet-18, a clean $\rightarrow$ corruption shift, and the two normalisations that govern the post-switch recovery. Table 6.3 reports the headline block at momentum 0.9, the practically relevant regime.

Both gates carry to the deeper backbone. The rot-MNIST result carries over and strengthens. Under batch norm the curriculum cuts depth from 11.8 to 4.8 pp and asymmetric PER to 5.9 pp; under group norm, where vanilla ER suffers a catastrophic 41.2 pp drop, the curriculum cuts it to 5.2 pp and asymmetric PER to 5.6 pp. Every gate-vs-vanilla depth and area contrast holds across all five seeds, at final accuracy within about 0.01 of vanilla and with worst-case retention sharply improved, min-ACC rising from 0.31 to 0.67 under group norm. On this backbone the transient stays open for hundreds of steps, so the area carries as much of the comparison as the depth (Figure 6.6): vanilla ER under group norm crashes T_0 almost to chance and crawls back over the whole recovery window, while both gates hold T_0 near its plateau through the switch. At the matched 5,000

buffer and momentum 0.9 both gates land in the same few-pp band, with the curriculum the cheaper of the two (below), consistent with the momentum regime of Section 6.4. The few-pp dip both gates keep under batch norm is boundary-localised and recovers within tens of steps, consistent with the running statistics re-adapting to the shifted inputs, a pathway no gate on the update touches (Appendix C.6).

Mechanism, generalisation, and cost. Three further checks are reported in full in Appendix C.8. A momentum cross on the deep backbone (Table C.9) turns out to be only weakly informative: at momentum 0 the ResNet under-converges in ten epochs (vanilla ACC 0.65, against 0.730 with momentum), and on that degraded baseline both gates improve with momentum by similar margins (28.6 $\rightarrow$ 5.2 pp for the curriculum, 22.6 $\rightarrow$ 5.6 pp for the directional gate), so the cross cannot separate the predicted composition from a general convergence gain; the vanilla depth null (40.5 $\rightarrow$ 41.2 pp) sits on the same degraded baseline. The momentum signatures therefore rest on the rot-MNIST results, where the MLP isolates them cleanly (Section 7.4). A generalisation block (Appendix C.7, Table C.8) shows the curriculum beating a paired vanilla control on two corruptions it was never tuned on (shot noise 46.3 $\rightarrow$ 7.3 pp, contrast 8.4 $\rightarrow$ 3.9 pp under group norm), with a single three-task group-norm sequence flagged as a reliability boundary of the adaptive ratio when two consecutive corruptions are near-identical and both gradient norms become small (Figure C.1; traced in

Norm	Method	gap_depth (pp)	gap_area_end	ACC	min-ACC
BatchNorm	G1: Vanilla ER	11.8 ± 2.5	17.5 ± 4.6	0.723 ± 0.008	0.644 ± 0.023
	G2: Curriculum ($\lambda_{\min}=0.20$)	4.8 ± 0.7	3.6 ± 2.2	0.722 ± 0.011	0.713 ± 0.017
	G3: Asym. PER ($\delta=0.1$)	5.9 ± 4.3	3.3 ± 3.0	0.715 ± 0.011	0.698 ± 0.023
GroupNorm	G4: Vanilla ER	41.2 ± 13.5	34.9 ± 23.7	0.730 ± 0.011	0.309 ± 0.111
	G5: Curriculum ($\lambda_{\min}=0.20$)	5.2 ± 2.1	1.7 ± 0.9	0.731 ± 0.020	0.670 ± 0.055
	G6: Asym. PER ($\delta=0.1$)	5.6 ± 1.1	4.8 ± 3.8	0.741 ± 0.009	0.673 ± 0.023

Table 6.3: CIFAR-10 headline block (ResNet-18, momentum 0.9, buffer 5,000, five seeds). Both gates cut gap depth and area far below vanilla ER at matched final accuracy and much higher worst-case retention (min-ACC), under both normalisations. Neither was given a CIFAR-specific sweep; both carry their rot-MNIST configurations.

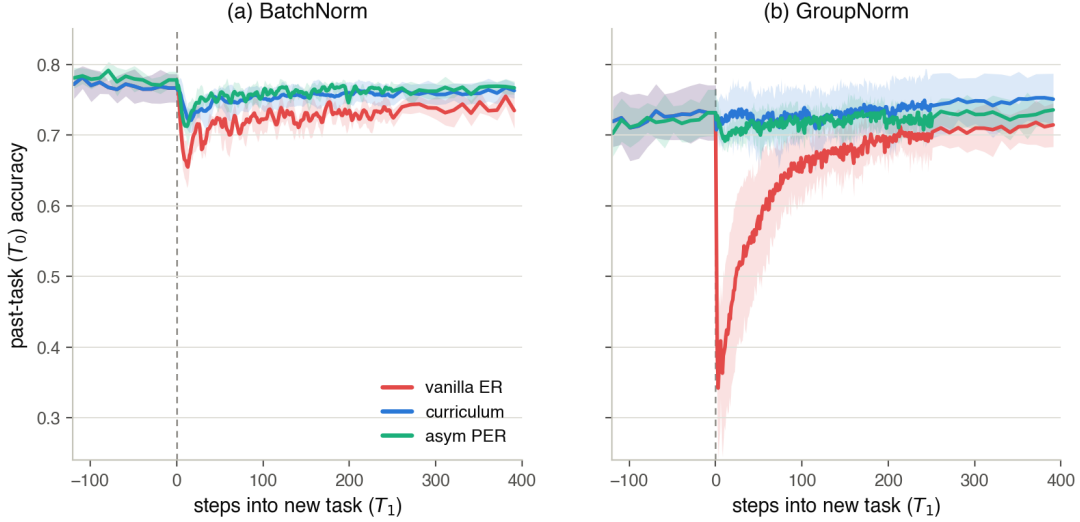


Figure 6.6: Per-step past-task (T_0) accuracy across the CIFAR-10 transition (ResNet-18, momentum 0.9, buffer 5,000), one panel per normalisation, each overlaying vanilla ER (red), the practical curriculum (blue) and asymmetric PER (aqua); seed mean with $\pm$ std bands, $x=0$ the task switch. (a) BatchNorm: vanilla ER dips and re-adapts slowly while both gates hold T_0 near its plateau. (b) GroupNorm: vanilla ER’s transient is deep and long, whereas both gates hold T_0 flat through the switch (Table 6.3, Appendix C.6).

Appendix C.7). Finally, compute separates the gates where retention does not (Table C.10): the scalar gate is one gradient-norm ratio on top of a standard ER step and leaves the GPU half-idle, while the directional gate’s 25 conjugate-gradient Fisher-vector products per step saturate it, costing $\sim 16\times$ the wall-clock and $\sim 24\times$ the GPU energy on the ResNet-18 at similar peak memory; on the MLP both runs finish in under half a minute and the difference is negligible. In the momentum regime standard in practice, then, the scalar gate matches the directional gate’s retention at essentially vanilla-ER cost and without the blind-spot cliffs, which is why it is the default recommendation.

6.6 Summary of Findings

The account holds end to end, with one revision it forced. The gap separates into overshoot of the followed path and a deterministic arc of the intended one (RQ1), and the step-size ladder separates them as a limit: at momentum 0 the excursion falls as $\eta \rightarrow 0$

but not to zero, converging on 3.8 pp of depth and 0.34 of flow-time area at ACC 0.898 on the exact replay gradient (4.7 pp / 0.27 / 0.877 on the sampled buffer), so between two thirds and three quarters of the depth measured at the standard operating point is discretisation overshoot and the remainder is the path itself. That remainder is the revision the measurement forced on the account’s metric mapping: depth is overshoot-dominated at the operating point but it is not overshoot alone. The arc is what the gates have to remove, and at the standard step size, where the ladder’s own remedy costs a hundredfold longer run. The scalar feedback gate with momentum reaches 3.7 pp of depth at vanilla accuracy, and C8 shows the pairing to be forced rather than fortunate, since the residual the schedule leaves at momentum 0 is estimator variance and momentum is the variance reducer a 1k memory can afford (RQ2); the directional feedforward gate shrinks depth and arc monotonically at no measured plasticity cost, but fails late and abruptly at its strongest diagnostic setting and re-inflates un-

der momentum (RQ3); the comparison is therefore regime-dependent, and both gates carry to CIFAR-10 at matched accuracy and much improved worst-case retention (RQ4). The discussion reads these findings against the account of Section 3.

7 Discussion

The results support the two-level account and, with it, the gate view of replay. This section reads them back against the theory, weighs the limits of the evidence, and draws out what follows for method design.

7.1 Why the Gap Needs Two Levels

The central claim is conceptual, and the experiments support it: the stability gap is not one quantity but two deviations of the same optimisation, kept apart because they answer to different interventions. The step-size ladder of Section 6.1 is the direct test, because a uniform rescaling of η is a time reparametrisation of the same gradient flow and is therefore the one intervention that changes how faithfully the intended trajectory is followed without changing which trajectory that is. Neither kind of intervention can substitute for the other, and the ladder makes one half of that statement a limit rather than a comparison: with both Level-Two drivers switched off at once, the step driven towards zero and the replay gradient exact, what is left is not zero but the arc. The converse holds at the standard operating point, where the curriculum replaces the path and the overshoot stays standing behind it; the transient only closes almost completely when the path repair is joined by the removal of the Level-Two driver it leaves untouched, the estimator variance of the finite buffer (C8, Section 6.2). The ladder also settles what the boundary imbalance is, which a driver reading would have placed on the overshoot side. Rescaling η leaves the vector field untouched, so the imbalance at θ_0^* is identical at every rung, the same vanishing replay gradient meeting the same large new-task gradient; yet the spike it produces at the standard step is gone at a third of that step, and what remains is smooth. The imbalance is Level-One terrain and the step size that meets it is the acute driver. At the same time, no intervention here is pure. The curriculum, nominally a path intervention, also shrinks the early push and with it the overshoot. An intervention aimed at one level can therefore move the other; what it cannot do is replace an intervention at the other level, and that non-substitutability is the practical content of the separation.

The natural alternative, a single additive decomposition of the gap into magnitude, variance, and trajectory shares, does not survive the data. The shares are order-dependent (a noisy replay gradient inflates the first-step magnitude), the “magnitude” share dissolves once the step size that converts it into a drop

is taken away, and the mapping from factors to metrics is crossed by the experiments, the curriculum’s momentum-0 tail looking like a “trajectory” term until C8 removes it with the exact buffer and no momentum at all, revealing it as variance (Table C.3). The two-level split is instead by kind: which path is followed versus how tightly it is followed.

The split also separates two costs that usually travel together. C8 closes the transient almost completely while its final accuracy sits well below vanilla ER’s (Section 6.2). The permanent rise of the past-task loss to its joint level is the stability–plasticity cost, and it remains after the transient is engineered away; the dip below that level and back, the part continual evaluation reveals, is a failure to follow the reference path, and the C8 cell shows it is removable. The gap is the second cost, not the first: it is not the price of learning the new task.

Finally, the two levels explain why the metric that matters changes with the backbone. On the MLP the network reconverges in a few steps, so the arc integrates to little and the perceived severity is the overshoot spike; on the ResNet the same transient stays open for hundreds of steps (Figure 6.6), so the slow tail is where most of the lost accuracy accumulates and where both gates’ margin over vanilla is largest (Table 6.3). Depth and area are the signatures of the two levels, and a depth-only reading would have understated the result on exactly the architectures where continual-evaluation cost is highest.

7.2 Two Gates, Two Ways to Divide the Work

In the unified update of Section 3.3, the curriculum and asymmetric PER occupy the taxonomy’s two pure corners: a scalar gate driven by feedback, and a directional gate driven by feedforward. The results show the two corners working and failing exactly where the taxonomy says they should, and the single fact that organises the whole comparison is *where* the gate attenuates the push.

The scalar feedback gate attenuates at source: it multiplies the entire new-task loss by a schedule that starts near zero, so early in the transition there is almost no push to discretise, accumulate, or let noise act on. Three consequences follow, and the data confirm each. It composes with momentum, because the accumulator never sees a sustained full-strength push to rebuild (Section 6.2). Read inside the schedule alone that composition would be indistinguishable from momentum’s general convergence gain; what turns the mechanistic reading into an observation is the contrast with the directional gate, which attenuates the same push at the step and whose protection the same accelerator undoes (Section 6.3). The two gates have little in common beyond the one-sided form and the gradient they attenuate, yet momentum moves them

in opposite directions, and the direction follows where the attenuation lands. It has no curvature blind spot, because the adaptive schedule reads a live damage signal rather than a model of the past task. And it pays a plasticity debt: holding the loss down slows the new task, a debt only acceleration repays (momentum restores C8’s final accuracy, Table C.3). The scalar gate’s price is plasticity, and momentum is how it is paid.

The directional feedforward gate attenuates per step: it filters the push through the replay Fisher, slowing the new task only along the directions the past-task curvature flags rather than scaling the whole loss down, and protects the past task at no measured plasticity cost, ACC and short-horizon plasticity rising as the filter strengthens (Table 6.2). The same per-step locality is its weakness on two axes, one the taxonomy predicts and one that emerged unanticipated. Under momentum the protection unravels by the re-inflation mechanism of Section 6.3, depth climbing from ≈ 2 to ≈ 8 pp: the same lever that closes the scalar gate’s gap re-opens the directional gate’s, and the difference is entirely where the attenuation is applied, at the source or at the step. That one distinction accounts for the whole momentum column of the study: the vanilla null, the curriculum’s composition, and the directional gate’s re-inflation.

The directional gate’s second failure was not predicted, and it is the sharper one because it makes the case for feedback. At the strongest filter the past-task curve can slide off a late cliff (the $\delta=0.03$ full-buffer cliffs of Section 6.3). A plausible reading is a blind spot of its curvature model: the sampled Fisher rates a direction flat that the true past-task loss punishes further out, and new-task pressure escapes through it. The exposure is characteristic of feedforward control, whose gate can only be as good as its model of the past task; a feedback gate instead reads the damage as it arrives, and its own weakness lies in the conditioning of the signal it reads, as the three-task sequence shows (Section 7.4). The cliffs and the plasticity debt are thus the two gates’ defining and complementary costs. The feedforward gate protects without holding the whole new task back, but it is only as good as its local curvature model of the past task and can fail where that model is wrong. The feedback gate cannot be fooled that way, because it reads the damage itself, but it pays by holding the whole new task back, a debt only an accelerator repays. This is why the two flip with momentum rather than sitting on a fixed frontier (Section 6.4). Both gates carry to the ResNet at matched accuracy and much improved worst-case retention (Table 6.3), though the momentum cross there is confounded (Section 7.4); it is the gate principle, not either particular method, that generalises.

7.3 What the Account Changes in the Literature

The two-level split lets the existing accounts of the gap compose rather than compete. De Lange et al. [4] trace the drop to the gradient imbalance at the boundary; Hess et al. [12] show the gap survives training on the exact joint loss, and Kao et al. [15] derive the geometry that makes it do so. Read through the split, these are observations about different parts of one mechanism, not rival explanations, and the ladder assigns each its place. The imbalance is a feature of the intended trajectory near the boundary, and by itself it does not produce the drop that is measured: it is the same at every rung, since rescaling the step size leaves the vector field untouched, yet the spike is already gone at a third of the standard step (Section 6.1). What converts the imbalanced field into that drop is the step size that meets it, which the ladder isolates by holding the field fixed and varying nothing else, and which Obis [22] reach from the optimiser side, where the drop’s depth and duration shift with the optimiser chosen. The persistence under exact gradients is the arc, and on the ladder’s exact arm it is measured rather than inferred, as the excursion that survives the exact gradient and the vanishing step together. None of the three accounts is displaced; the decomposition connects the imbalance account to the measurements through the step size, and situates the geometry account as the part no repair of the step reaches.

The unified update of Sodhani et al. [25] sorts replay methods by which weight they move and from what signal; the results add the axis that turned out to be decisive, where the modulation is applied. AGEM and MEGA hold the new-task weight at one and correct per step from a live signal [2, 9], which places them on the step side of the source/step divide, the side whose protection momentum unwinds in Section 6.3. The account therefore predicts that their boundary protection weakens under momentum as the directional gate’s does, while source-attenuating schedules compose with it. These results do not test that prediction, but it is directly testable, and it matters because the original comparisons predate the continual-evaluation lens: end-of-task metrics would not have registered per-step protection eroding, any more than they register the gap itself.

Mode connectivity enters the account as an existence result and exits as a control result. Garipov et al. [7] and, for the continual setting, Mirzadeh et al. [21] establish post hoc that the relevant solutions share a low-loss corridor; the corridor is inspected after training and says nothing about whether an optimiser can be kept inside it while learning. C8 supplies the dynamic counterpart: an update rule whose measured past-task accuracy stays essentially flat through the transition, consistent with the iterate staying near the

monotone reference path, turning the observation that a low-loss route exists into the statement that a gate can keep training close to it. The same experiment sharpens the stability–plasticity dilemma [20] into a measurement: the dilemma’s price is the permanent offset to the joint level, and it bounds where the model can settle, not how it gets there; the deep boundary transient that continual evaluation reveals is a routing failure the right gate removes.

7.4 Limitations

Several limits bound these claims, beginning with the statistical one: with five seeds the design cannot reach a 0.05 threshold (Section 5.1.2), so the argument rests on effect sizes and seed-paired direction. The effects the story leans on, the up-to-eightfold depth reductions on CIFAR, the C7→C8 variance collapse, the momentum re-inflation of the directional gate, are large relative to the seed spread and unanimous across seeds; the smaller ones, the adjacent- δ steps and the $\lambda_{\min}$ trade-offs, are read as directional consistency only, and one, the adaptive-versus-best-linear depth, does not reach even that ($p = 0.44$).

What the ladder settles at momentum 0 it settles under momentum too, but the two answers are not directly comparable. Momentum places its leg a decade coarser in effective step, so only the fourth rung reaches the range in which the momentum-0 leg had already converged, and there it does flatten onto an arc of its own (Section 6.1). Because each leg warm-starts from an anchor trained at its own momentum and the two settle some seven accuracy points apart, the difference between their limits is a difference between two operating points rather than a measurement of what momentum does to the arc; isolating that would need a shared anchor this design does not provide. The arc’s magnitude is also one measurement in one place, two-task rot-MNIST on an MLP from a single anchor θ_0^* per momentum setting and seed. That the excursion converges on a positive limit is a structural claim the design supports; how large the arc is on another benchmark or backbone these runs cannot say, and the ResNet’s far longer transient (Figure 6.6) suggests the balance between the two levels is itself architecture-dependent. Nor is the ladder’s own remedy a method. Matched flow time makes the epoch count scale as $1/\eta$, so the rung at which the momentum-0 leg converges costs ten times the training and the momentum leg needs a further decade below it; the shorter step buys the separation and is a diagnostic, not a technique to be adopted in place of a gate.

The gates carry limits of their own. The scalar gate’s accuracy-neutrality is a property of the full curriculum-plus-momentum configuration rather than of the ramp: at momentum 0, lengthening the linear ramp lowers depth but costs several points of fi-

nal accuracy (Table 6.1). Its feedback signal has a reliability boundary, the three-task group-norm sequence in which the adaptive ratio loses conditioning at a transition between two near-identical corruptions and the schedule underperforms its control with high seed variance; we flag that as a property of the gradient-ratio signal, absent from the batch-norm variant, rather than of the gate principle, and Appendix C.7 traces it step by step through the logged norms. For the directional gate the solver control establishes that the cliffs are real and not a numerical artefact, but the blind-spot mechanism we offer for them is a reading consistent with the data, not an established cause.

The deep-backbone evidence is thinner in one specific place. The CIFAR momentum cross is uninformative in both directions: at momentum 0 the ResNet under-converges, both gates improve with momentum by similar margins, and the vanilla null sits on that same degraded baseline, so no cell of it cleanly confirms or refutes a momentum signature. All three momentum signatures therefore rest on the rot-MNIST results; a clean deep-backbone test would require a converged momentum-0 baseline we did not run.

Two scope limits are worth stating plainly. The exact-gradient conditions (the ladder’s 60k arm, C8, and the full-buffer leg of the δ sweep) store the entire past task and so break the limited-memory premise of continual learning; they are diagnostic probes that isolate the arc and split momentum’s roles, and the practical configurations are the 1k-buffer curriculum and the standard-buffer directional gate. And the directional gate’s per-step Fisher-vector solve adds a fixed multiple of the backward cost per step, $\sim 16\times$ the wall-clock and $\sim 24\times$ the GPU energy of the curriculum on the ResNet-18 (Table C.10), an overhead the scalar gate’s single gradient-norm ratio does not carry. The headline analysis also rests on a single-transition, two-task design chosen to isolate one boundary; the five-task rot-MNIST sequence (Appendix C.9) shows the regime picture surviving four consecutive boundaries, but a systematic study of longer sequences and of class-incremental rather than domain-incremental shifts is left open.

7.5 Outlook

The taxonomy that organises the two methods also names what is missing from it. The gates studied here fill three of its four corners: scalar $\times$ feedback, directional $\times$ feedforward, and, as an ablation, the linear ramp at scalar $\times$ feedforward. The remaining corner, a directional gate driven by a live damage signal rather than by a precomputed curvature model, is the natural candidate the two failure modes point to: it would filter the push by direction, keeping the plasticity that makes the feedforward gate cheap, while reading the damage as it arrives rather than trust-

ing a model of the past task. Whether one design can deliver both properties at once is an open question, not an implication of these results, but the momentum results constrain any attempt in advance: its directional attenuation should act at source or on the accumulated velocity, not on the instantaneous step, or it inherits the re-inflation of Section 6.3.

Four narrower directions follow from the failure modes and costs. The first comes from the step-size results. The ladder has mapped the uniform step-size axis and mapped it to a limit, so what stays open on that axis is the non-uniform case: where per-parameter adaptive step sizes sit in the gate taxonomy [22], and whether an attenuation that is directional in the coordinate basis but applied to the step inherits the re-inflation that the directional gate suffers under momentum. Both are directly testable, and the second is the sharper question, because these optimisers carry an accumulator of their own. Second, the same repair applies to existing methods: any approach that preconditions the update with past-task curvature per step, the natural-gradient family included [15], carries the re-inflation exposure, and moving its attenuation to the accumulated velocity is a directly testable fix. Third, the directional gate’s compute cost is not fundamental: the per-step conjugate-gradient solve could be replaced by a diagonal or Kronecker-factored Fisher approximation, and the Fisher itself could be refreshed at coarser intervals, up to once per task boundary as Natural Continual Learning does [15]; whether the protection survives the coarser metric or the staler one is open; on backbones much larger than the ResNet-18 here, such an approximation is a precondition for the directional corner rather than a refinement of it. Fourth, momentum’s gap-closing role proved interchangeable with source variance reduction (C8), so the curriculum could be paired with other variance reducers; iterate averaging, which carries none of momentum’s accumulator-driven overshoot, is the natural candidate.

The cliffs add a methodological rule. They arrive late and abruptly, after long stretches of exemplary retention, and end-of-task metrics would have recorded the $\delta=0.03$ runs as successes; only per-step measurement caught the failure. Continual evaluation [4] is usually presented as the lens that reveals the stability gap. These results argue for it also as an acceptance test for any feedforward gate deployed in a system that serves predictions while it trains: a gate that acts on a model of the past task can fail precisely where that model is wrong, silently until it does.

Statement on the Use of AI Tools

In accordance with the University of Groningen’s guidelines on the responsible use of generative AI, I disclose that AI assistants were used during the prepa-

ration of this thesis. The tools were Google’s Gemini 3.x series, Anthropic’s Claude 4.x series, and Anthropic’s Claude Fable 5, the releases current across the period, February to July 2026, in which the work was carried out. They functioned as assistive tools under my continuous direction, not as autonomous contributors. Specifically, I used them to brainstorm and stress-test ideas and as a sounding board for reasoning through arguments and explanations; to draft and revise prose, which I subsequently edited for accuracy, precision, and voice; to assist in writing and refactoring the analysis and experiment code, all of which I reviewed, ran, and tested; and to produce first-pass summaries of the literature, which I verified against the primary sources.

The tools were never given open-ended control: every request specified a concrete, bounded task, and all output was reviewed and, where retained, corrected, understood, and integrated by me. All experimental results, figures, and numerical claims were checked against the underlying run data, and all cited work was read in the original. The research questions, experimental design, interpretation of the results, and the overall argument of the thesis are my own, and I take full responsibility for its content, including any errors.

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A Implementation Details

This appendix records the implementation specifics that the methods of Section 5 depend on but that would interrupt the argument in the main text: the datasets, the backbones and training protocol, the experience-replay variants, and the asymmetric-PER solver. The full implementation is available at <https://github.com/Arthurreuss/ValleyWalk>.

A.1 Datasets

rot-MNIST rotates each 28×28 image by the per-task angle and feeds the flattened greyscale vector to the MLP. The CIFAR-10 domain shifts are generated with the `imagecorruptions` package, which implements the standard common-corruptions benchmark of Hendrycks and Dietterich [11]; the Gaussian noise, shot noise, and contrast shifts are applied at severity 3 on the benchmark’s 1–5 scale, per sample at load time, followed by the standard CIFAR-10 channel normalisation. The clean task (“none”) applies the normalisation only.

A.2 Models and Training

Tables A.1 and A.2 give the two backbones, and Table A.3 the training protocol shared across both benchmarks.

Component	Value
Input	784 (flattened 28×28 greyscale)
Hidden layers	2×400 (fully connected, ReLU)
Output	10-class logits
Regularisation	none
Total parameters	478,410

Table A.1: MLP backbone used for the rot-MNIST sweeps.

Component	Value
Input	$3 \times 32 \times 32$ RGB
Stem	3×3 conv, stride 1, no max-pool (CIFAR adaptation)
Stages	4 residual stages, BasicBlocks [2, 2, 2, 2], widths [64, 128, 256, 512]
Normalisation	BatchNorm (default); GroupNorm ($\min(32, C)$ groups) variant crossed in Section 5.2.5
Head	global average pool + linear, 10-class logits
Total parameters	≈ 11.2 M

Table A.2: CIFAR-adapted ResNet-18 backbone used for the CIFAR-10 block. The 3×3 stride-1 stem and removed max-pool keep the feature map at 32×32 ; all other details follow the ResNet-18.

Hyperparameter	rot-MNIST (MLP)	CIFAR-10 (ResNet-18)
Epochs per task	1 (online); c in the ladder	10
Batch size	256	256
Optimiser	SGD	SGD
Learning rate	0.1; $0.1/c$, $c \in \{1, 10, 33\}$, in the ladder	0.1
Momentum	0.0 or 0.9	0.9 (0.0 in cross)
Weight decay	0.0	0.0
Replay buffer	1,000 or 60,000	5,000
Seeds	5	5 or 3

Table A.3: Training protocol for both benchmarks.

A.3 Experience replay

The replay buffer uses reservoir sampling [29], so the stored set is a uniform sample of the past stream at any budget. Two modes appear in the experiments:

- *Standard ER* (the D1 reference, the ladder’s 1k arm, and all C-series conditions except C8): each step takes one combined backward pass on $L_{\text{current}} + L_{\text{replay}}$, with half the mini-batch drawn from the current task and half sampled with replacement from the buffer.
- *Full-data replay* (the ladder’s exact arm, C8, and the P2–P3 legs): the replay gradient is computed on *every* stored sample exactly once per step (no sub-sampling). Paired with a buffer equal to the full past-task training set (60,000 for rot-MNIST T_0), this yields the deterministic empirical past-task gradient at the current parameters, removing estimator variance.

The λ -curriculum re-weights only the current-task term, minimising $L_\lambda = L_{\text{replay}} + \lambda(t) L_{\text{current}}$; the adaptive schedule reads $\|g_{\text{replay}}\|/\|g_{\text{new}}\|$ through an exponential moving average ($\alpha = 0.05$) and applies the optional floor λ_{min} .

A.4 The step-size ladder

Every cell of the ladder warm-starts from a shared task-0 checkpoint. `init.checkpoint` loads a task-0 `state.dict` written by a separate anchor run, one per momentum setting and seed, trained at the standard operating point ($\eta = 0.1$, one epoch); `init.skip_tasks` then suppresses the training loop of that many leading tasks while still running their `end.task` bookkeeping, so the replay buffer is filled from the skipped task’s stream exactly as it would have been in an ordinary run. Because reservoir sampling at a fixed seed returns the same subset of the same stream at a given budget, θ_0^* and the buffer are identical across the three rungs at a given buffer arm, momentum, and seed, and the learning rate is the only difference between them.

A skipped task still advances `global_step` by one nominal epoch ($\lceil 60,000/256 \rceil = 235$ batches), so the task boundary carries the same index as in an ordinary run and the per-step plotting convention of Section 5.1.1 keys off it unchanged. One pre-switch evaluation is written at step 234, supplying the flat pre-boundary segment that the suppressed training loop would otherwise have recorded.

Inside the dense post-switch window the stability-gap cadence fires on the step index *within the current task* rather than on the global step. At a cadence of $c > 1$ a global-step modulus would place the first post-switch sample at an arbitrary phase offset into the task whenever c does not divide the 235-batch epoch, skipping exactly the steps the boundary spike occupies. The change is a no-op at cadence 1, which is every previously archived run.

The four rungs use $\eta = 0.1$, $\eta = 0.1/3 = 0.0\bar{3}$, $\eta = 0.01$ and $\eta = 0.001$. The second is written into the configuration at full precision rather than rounded, so that $\eta c = 0.1$ holds exactly and the flow time $\tau = 23.5$ is matched to the last digit; the main text reports it as 0.1/3. The dense window is set to $250c$ steps, which exceeds the $235c$ steps of the task, so every cell records the whole of T_1 at the dense cadence: 235 samples per rung, evenly spaced in flow time. Buffer-fidelity diagnostics are disabled throughout the ladder, since each such step costs a full pass over the past-task training set and the longest rung takes thirty-three epochs.

A.5 Asymmetric PER

How the Fisher is computed. For a softmax cross-entropy model the Fisher information matrix equals the generalised Gauss–Newton matrix

$$F = \frac{1}{B} \sum_b J_b^\top H_b J_b, \quad J_b = \frac{\partial z_b}{\partial \theta}, \quad H_b = \text{diag}(p_b) - p_b p_b^\top,$$

where z_b are the logits of sample b and $p_b = \text{softmax}(z_b)$ [18]. This matrix is positive semi-definite by construction, so $F + \delta I$ is positive-definite and the solve is well-posed; it depends only on the model’s predictive distribution, not on the labels. F_{rep} is estimated on a mini-batch sampled from the replay buffer and *recomputed at every step from the current parameters*: the gate uses a live local curvature model, not a snapshot frozen at the task boundary. It remains feedforward in the sense of Section 3.3, because what it reads is the curvature model, never the past-task performance it is meant to protect. The matrix is never formed explicitly. Fisher-vector products $v \mapsto Fv$ are computed matrix-free by the Gauss–Newton trick of Schraudolph [24]: one forward pass gives the logits (retained across the solve), each product then costs one Jacobian-vector and one vector-Jacobian product.

The solve. The filtered push $\delta(F_{\text{rep}} + \delta I)^{-1} g_{\text{cur}}$ of Equation (3.4) is computed each step by conjugate gradients over these Fisher-vector products (25 iterations, relative-residual tolerance 10^{-4}), warm-started from the previous step’s solution, which keeps the per-step solve cheap across the run; the warm start is dropped at task

boundaries, where the landscape shifts. The solver control of Section 5.2.4 reruns the strongest filter ($\delta = 0.03$, full buffer) at 60 iterations to separate properties of the gate from artefacts of an under-converged solve. On the full-buffer leg the replay gradient is exact over all 60k past samples while the Fisher stays estimated on a sampled batch, since a full-buffer Fisher-vector product per CG iteration would dominate the runtime.

Order of gate and momentum. When momentum is enabled, the gate is applied first: the filtered direction d is written in place of the gradient, and the optimiser’s velocity buffer accumulates $v \leftarrow \mu v + d$ with the step $-\eta v$. The velocity is therefore built from already-filtered directions and is itself applied unfiltered. The alternative ordering, accumulating raw gradients and filtering the velocity, is a distinct design with potentially different behaviour under the re-inflation mechanism of Section 6.3; it is not implemented here and is discussed as future work in Section 7.5.

B Metric Definitions

This appendix gives the full definitions of every metric recorded by the evaluation pipeline of Section 5.1.2. The main text explains the measures the argument turns on; the tables below are the look-up reference for the complete set.

Stability-gap metrics. Computed from the dense per-step evaluation. Write a_j^{pre} for the pre-task accuracy on task j and $a_j(t)$ for its accuracy at step t of the new task.

Metric	Definition	Role
gap_depth	$\max_j (a_j^{\text{pre}} - \min_t a_j(t))$ over the full record window	Primary metric, worst single point of instability
gap_area_end	$\sum_j \int_0^T \max(0, b_j - a_j(t)) dt$, trapezoidal over the full new-task record (to task end T), with $b_j = \min(a_j^{\text{pre}}, \bar{a}_j^{\text{end}})$ and $\bar{a}_j^{\text{end}}$ the trailing-window mean (last $\max(5, \lceil 0.1n \rceil)$ samples); accuracy at or above b_j contributes 0	Transient cost reported in the results: dip below the recovered level until it closes, excluding permanent forgetting and continued learning (units: accuracy · steps)
recovery_steps	The first step (counted from the transition) at which <i>every</i> past task is simultaneously back at $\geq 90\%$ of its pre-task baseline; undefined (reported as “—”) if no sub-threshold dip occurs, or if recovery is not reached within the record	Duration of the transient, how long the gap stays open

Table B.1: Stability-gap metrics, recorded from dense per-step evaluation. `gap_depth` and `gap_area_end` are the two axes used in the main analysis, how deep the gap cuts and how long it stays open, together with `recovery_steps`.

Continual-learning metrics. Following De Lange et al. [4], computed at task boundaries and over the full record. Let $A(E_j, f_n)$ be the accuracy on the test set of task j evaluated at the model f_n after training stage n , and let N be the number of tasks. The full task-boundary accuracy matrix $R[i, j] = A(E_j, f_{T_i})$ is logged for every run, so any downstream metric can be reconstructed.

Metric	Definition	Role
ACC	$\frac{1}{N} \sum_j A(E_j, f_{T_{N-1}})$	Average task accuracy at the final model
FORG	$\frac{1}{N-1} \sum_{i < N-1} [A(E_i, f_{T_i}) - A(E_i, f_{T_{N-1}})]$	Average drop from right-after-learning to final
min-ACC	$\frac{1}{N-1} \sum_{i < N-1} \min\{A(E_i, f_n) : n > T_i\}$	Worst-case retention since learning
WF _w	Mean over tasks of the largest accuracy drop in any window of w consecutive evaluations ($w \in \{10, 100\}$)	Worst-case stability over short / medium horizons
WP _w	Symmetric counterpart of WF _w : the largest accuracy gain ($w \in \{10, 100\}$)	Plasticity / learning velocity
WC-ACC	$\frac{1}{N} A(E_{N-1}, f_{T_{N-1}}) + (1 - \frac{1}{N}) \text{min-ACC}$	Worst-case trade-off, a lower bound on ACC

Table B.2: Continual-learning metrics of De Lange et al. [4], recorded at task boundaries and over the full record.

Gradient-fidelity diagnostics. Recorded at the dense cadence for the decomposition conditions only. The tracker compares the replay-buffer gradient g_{replay} against the true past-task gradient g_{true} , computed at every step by a separate forward and backward pass over the entire past-task training set (no sub-sampling, so g_{true} is the deterministic empirical past-task gradient at the current parameters).

Diagnostic	Logged name	Role
$\cos(g_{\text{replay}}, g_{\text{true}})$	<code>true_grad_cosine</code>	Buffer-side directional fidelity, the estimator-variance driver of Section 5.2.1
$\ g_{\text{replay}}\ / \ g_{\text{true}}\ $	<code>true_grad_mag_ratio</code>	Buffer-side magnitude fidelity
$\ g_{\text{new}}\ / \ g_{\text{replay}}\ $	<code>grad_ratio</code>	Update-side magnitude asymmetry, large at step 1 when $\ g_{\text{replay}}\ \approx 0$

Table B.3: Gradient-fidelity diagnostics. Each is logged as a per-step series together with its mean (and minimum, for the cosine). The first two carry the `true_grad` prefix because they are measured against g_{true} ; `grad_ratio` compares g_{replay} to the current-task gradient g_{new} and its inverse drives the adaptive curriculum of Section 5.2.2.

C Complete Results

This appendix gives the full metric set for every condition of the main experimental blocks, as recorded by the aggregation pipeline of Section 5.1.2. Each row is the mean over five seeds (three for the CIFAR generalisation block) with the seed standard deviation. Accuracy-type metrics (ACC, FORG, min-ACC, WF₁₀₀, WP₁₀₀, WC-ACC) are fractions; `gap_depth` is in percentage points (pp); `gap_area_end` is the transient cost (accuracy-fraction $\times$ steps, Table B.1), the dip below the level each task recovers to, so permanent forgetting does not contribute; “Recov.” is `recovery_steps`, with “—” marking conditions where every past task stayed above the 90% recovery threshold for the whole window (no recovery event to time).

C.1 rot-MNIST: The Step-Size Ladder (S-series)

The sixteen cells of the ladder of Section 5.2.1: four rungs at matched flow time $\tau = 23.5$, each on the 1k reservoir and on the 60k exact past-task gradient, at both momentum settings, all warm-started from one frozen θ_0^* per momentum setting and seed.

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	A_τ
<i>Momentum 0.0</i>								
S1 $\eta=0.1$ (1k)	0.874 ± 0.008	0.020 ± 0.019	0.653 ± 0.035	0.153 ± 0.017	0.454 ± 0.020	0.770 ± 0.019	22.9 ± 4.2	0.577 ± 0.080
S1 $\eta=0.1$ (exact)	0.895 ± 0.002	-0.020 ± 0.014	0.658 ± 0.044	0.138 ± 0.026	0.453 ± 0.025	0.773 ± 0.022	22.3 ± 5.0	0.590 ± 0.172
S2 $\eta=0.1/3$ (1k)	0.876 ± 0.004	0.021 ± 0.012	0.826 ± 0.007	0.036 ± 0.004	0.365 ± 0.005	0.858 ± 0.004	5.6 ± 1.6	0.293 ± 0.044
S2 $\eta=0.1/3$ (exact)	0.897 ± 0.002	-0.021 ± 0.012	0.839 ± 0.002	0.029 ± 0.002	0.369 ± 0.006	0.865 ± 0.002	4.2 ± 1.2	0.360 ± 0.167
S3 $\eta=0.01$ (1k)	0.877 ± 0.002	0.020 ± 0.013	0.832 ± 0.006	0.031 ± 0.004	0.368 ± 0.005	0.862 ± 0.003	4.9 ± 1.4	0.275 ± 0.055
S3 $\eta=0.01$ (exact)	0.897 ± 0.002	-0.020 ± 0.012	0.842 ± 0.003	0.025 ± 0.002	0.374 ± 0.003	0.867 ± 0.002	3.9 ± 1.2	0.346 ± 0.164
S4 $\eta=0.001$ (1k)	0.877 ± 0.002	0.020 ± 0.013	0.834 ± 0.006	0.028 ± 0.004	0.369 ± 0.006	0.863 ± 0.003	4.7 ± 1.4	0.270 ± 0.052
S4 $\eta=0.001$ (exact)	0.898 ± 0.001	-0.022 ± 0.012	0.843 ± 0.003	0.023 ± 0.002	0.376 ± 0.003	0.868 ± 0.002	3.8 ± 1.2	0.341 ± 0.163
<i>Momentum 0.9</i>								
S1 _M $\eta=0.1$ (1k)	0.929 ± 0.003	0.058 ± 0.006	0.802 ± 0.023	0.087 ± 0.008	0.442 ± 0.006	0.881 ± 0.011	15.4 ± 2.2	0.105 ± 0.022
S1 _M $\eta=0.1$ (exact)	0.965 ± 0.001	-0.011 ± 0.004	0.814 ± 0.011	0.076 ± 0.006	0.457 ± 0.009	0.889 ± 0.006	14.1 ± 1.2	0.301 ± 0.034
S2 _M $\eta=0.1/3$ (1k)	0.936 ± 0.003	0.053 ± 0.003	0.855 ± 0.009	0.054 ± 0.004	0.436 ± 0.008	0.913 ± 0.004	10.1 ± 0.7	0.021 ± 0.005
S2 _M $\eta=0.1/3$ (exact)	0.970 ± 0.001	-0.015 ± 0.003	0.857 ± 0.005	0.053 ± 0.002	0.454 ± 0.007	0.913 ± 0.003	9.9 ± 0.5	0.122 ± 0.014
S3 _M $\eta=0.01$ (1k)	0.940 ± 0.002	0.048 ± 0.003	0.901 ± 0.005	0.029 ± 0.002	0.419 ± 0.006	0.937 ± 0.003	5.4 ± 0.3	0.003 ± 0.001
S3 _M $\eta=0.01$ (exact)	0.972 ± 0.001	-0.017 ± 0.003	0.905 ± 0.005	0.027 ± 0.001	0.435 ± 0.007	0.939 ± 0.003	5.0 ± 0.3	0.061 ± 0.012
S4 _M $\eta=0.001$ (1k)	0.942 ± 0.002	0.045 ± 0.002	0.910 ± 0.003	0.020 ± 0.001	0.411 ± 0.007	0.942 ± 0.002	4.6 ± 0.2	0.0006 ± 0.0001
S4 _M $\eta=0.001$ (exact)	0.973 ± 0.001	-0.018 ± 0.003	0.928 ± 0.003	0.014 ± 0.002	0.426 ± 0.007	0.950 ± 0.002	2.7 ± 0.4	0.040 ± 0.011

Table C.1: Step-size ladder on two-task rot-MNIST, full metrics: rungs $\eta = 0.1, 0.1/3, 0.01$ and 0.001 at 1, 3, 10 and 100 epochs, each on both buffer arms and at both momentum settings, five seeds. Depth is a maximum over accuracies and so is η -invariant. Area is *not*: gap_area_end is a trapezoidal integral of the drop curve against the step index, weighted by the spacing between records (units accuracy-steps, Section 5.1.2), so an identically shaped excursion reads $c = 0.1/\eta$ times larger on a rung stretched by c . The area column is therefore reported in flow time, $A_\tau = \eta A_{\text{steps}}$, which is the integral of the same curve against $\tau = \sum_t \eta$ and is comparable across rungs; the fixed- η blocks elsewhere in this appendix keep the step-axis value. WF₁₀₀ and WP₁₀₀ are fixed hundred-step horizons and therefore span a hundredth of the flow time at $\eta = 0.001$ that they span at $\eta = 0.1$; their movement down the ladder is a units artefact of the reparametrisation (Section 6.1).

C.2 rot-MNIST: λ -Curriculum (C-series)

Vanilla ER at the standard protocol (D1: 1k reservoir, $\eta = 0.1$, one online epoch per task, the two gradients summed as they come) is the reference every rot-MNIST comparison is quoted against, and its full metric row heads each momentum block below rather than standing in a table of its own. These are the runs from which the buffer-fidelity diagnostics of Table C.4 (Appendix C.3) are taken; the step-size ladder re-runs its own $\eta = 0.1$ rung rather than reusing the row, because every cell of the ladder has to share one frozen θ_0^* (Appendix A.4).

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end	Recov.
<i>Momentum 0.0</i>									
Vanilla ER (baseline)	0.873 ± 0.005	0.019 ± 0.020	0.706 ± 0.056	0.147 ± 0.029	0.736 ± 0.020	0.794 ± 0.028	19.5 ± 4.1	6.3 ± 0.5	3.6 ± 0.9
C1 linear N50	0.851 ± 0.023	0.042 ± 0.021	0.707 ± 0.019	0.148 ± 0.011	0.748 ± 0.006	0.784 ± 0.021	17.5 ± 1.7	4.9 ± 0.5	35.8 ± 6.3
C2 linear N100	0.844 ± 0.025	0.046 ± 0.023	0.719 ± 0.033	0.140 ± 0.025	0.730 ± 0.006	0.786 ± 0.019	16.2 ± 4.3	3.6 ± 0.3	64.0 ± 8.6
C3 linear N200	0.828 ± 0.025	0.054 ± 0.024	0.739 ± 0.048	0.128 ± 0.043	0.688 ± 0.004	0.784 ± 0.034	14.3 ± 4.6	1.8 ± 0.6	132.8 ± 43.4
C4 adaptive	0.833 ± 0.025	0.050 ± 0.019	0.754 ± 0.025	0.111 ± 0.016	0.703 ± 0.007	0.794 ± 0.020	12.8 ± 2.9	2.2 ± 0.5	88.8 ± 6.1
C5 adaptive lmin0.05	0.835 ± 0.025	0.048 ± 0.019	0.749 ± 0.027	0.113 ± 0.016	0.702 ± 0.006	0.793 ± 0.020	13.2 ± 3.2	2.2 ± 0.5	88.8 ± 6.1
C6 adaptive lmin0.10	0.836 ± 0.026	0.048 ± 0.021	0.743 ± 0.026	0.120 ± 0.019	0.700 ± 0.007	0.791 ± 0.019	13.9 ± 3.3	2.4 ± 0.5	88.4 ± 11.0
C7 adaptive lmin0.20	0.837 ± 0.031	0.049 ± 0.024	0.743 ± 0.030	0.119 ± 0.023	0.710 ± 0.010	0.792 ± 0.024	13.9 ± 3.4	2.7 ± 0.6	64.8 ± 8.2
<i>Momentum 0.9</i>									
Vanilla ER (baseline)	0.928 ± 0.003	0.058 ± 0.007	0.797 ± 0.028	0.092 ± 0.016	0.808 ± 0.012	0.878 ± 0.014	15.9 ± 2.6	1.0 ± 0.2	12.6 ± 1.7
C1 linear N50	0.932 ± 0.003	0.053 ± 0.005	0.892 ± 0.004	0.041 ± 0.004	0.830 ± 0.012	0.927 ± 0.003	6.4 ± 0.5	0.2 ± 0.1	—
C2 linear N100	0.932 ± 0.003	0.051 ± 0.008	0.892 ± 0.005	0.036 ± 0.003	0.826 ± 0.012	0.926 ± 0.002	6.4 ± 0.6	0.2 ± 0.1	—
C3 linear N200	0.925 ± 0.007	0.051 ± 0.006	0.893 ± 0.004	0.031 ± 0.004	0.819 ± 0.010	0.920 ± 0.006	6.2 ± 0.6	0.1 ± 0.0	—
C4 adaptive	0.917 ± 0.001	0.019 ± 0.004	0.931 ± 0.005	0.016 ± 0.003	0.798 ± 0.010	0.915 ± 0.001	2.5 ± 0.5	0.3 ± 0.1	—
C5 adaptive lmin0.05	0.922 ± 0.001	0.023 ± 0.004	0.930 ± 0.005	0.016 ± 0.002	0.802 ± 0.009	0.921 ± 0.001	2.6 ± 0.5	0.1 ± 0.0	—
C6 adaptive lmin0.10	0.927 ± 0.002	0.028 ± 0.005	0.926 ± 0.004	0.017 ± 0.003	0.806 ± 0.009	0.927 ± 0.002	3.0 ± 0.4	0.0 ± 0.0	—
C7 adaptive lmin0.20	0.931 ± 0.002	0.035 ± 0.005	0.919 ± 0.003	0.021 ± 0.004	0.812 ± 0.009	0.930 ± 0.001	3.7 ± 0.4	0.0 ± 0.0	—

Table C.2: rot-MNIST λ -curriculum sweep, full metrics, at both momentum settings, each block headed by the vanilla-ER reference the sweep is quoted against. All conditions applied on top of standard ER (1k reservoir). Area is on the step axis, as in every block at fixed η ; the ladder converts to flow time (Table C.1).

The ingredient sweep: adaptive signal and $\lambda_{\min}$ floor. Table C.2 settles the two design choices of the shipped configuration, and both are decided in the momentum-on leg. At momentum 0 the adaptive schedule is indistinguishable from the longest linear ramp (12.8 pp / 2.2 / 0.833 against C3's 14.3 pp / 1.8 / 0.828, seeds split on depth, $p = 0.44$), so at that setting it is preferred for having no ramp length to tune rather than for its numbers. Under momentum the same pair separates: C4_M reaches 2.5 pp / 0.3 / 0.917 against C3_M's 6.2 pp / 0.1 / 0.925, shallower in all five seeds by ≈ 3.7 pp, so the feedback signal both removes the knob and beats the best setting of the knob, at a small cost in final accuracy and in the tail. The floor trades that cost back. Raising $\lambda_{\min}$ from 0 to 0.20 (C4_M $\rightarrow$ C7_M) moves depth from 2.5 to 3.7 pp, worse in all five seeds, while accuracy moves from 0.917 to 0.931 and the area from 0.3 to 0.03, better in all five seeds on both; the

intermediate settings $C5_M$ and $C6_M$ walk monotonically between the two ends in depth and accuracy, so the floor is a graded Pareto knob rather than a switch. $C7_M$ is the shipped point because it clears the tuned linear ramp on all three quantities (depth in five seeds of five, accuracy in four) and returns accuracy to vanilla ER’s level (3.7 pp / 0.03 / 0.931 against 15.9 pp / 1.0 / 0.928, above vanilla in four of five seeds and within the seed spread). At momentum 0 the floor is not a useful knob: it costs depth (12.8 $\rightarrow$ 13.9 pp) and returns almost nothing in accuracy (0.833 $\rightarrow$ 0.837), since early T_1 progress there is set by the boundary gradient the floor is too small to redirect.

Full-buffer momentum diagnostic (C8). Table C.3 crosses the practical curriculum with the full 60k buffer at both momentum settings, isolating momentum’s two roles (Section 6.2). Source variance reduction alone (C8, full buffer, $\mu=0$) already closes the gap to 0.2 pp, so the residual the curriculum leaves at $\mu=0$ is estimator variance, not structure; momentum’s gap-closing role is interchangeable with the full buffer. Acceleration alone (the $\mu=0.9$ column) restores accuracy: $C8_M$ reaches 0.955, approaching the exact-gradient ceiling (the ladder’s ungated $\eta = 0.1$ rung on the same 60k gradient at $\mu=0.9$: 0.965, Table C.1), while re-introducing a small transient (2.0 pp, area 0.38). The shape and timing of this transient point to the $\lambda_{\min}$ floor meeting momentum. With the replay gradient exact and the arc removed, the only push left in the first steps is the floored fifth of the new-task gradient (the logged gradient-norm ratio confirms the schedule sits at $\lambda = 0.2$ through the dip window), and momentum amplifies a persistent push toward $1/(1 - \mu) = 10$ times its per-step size: the dip accordingly opens at the boundary, bottoms out about ten steps in, and closes within about fifty as the replay gradient reasserts itself. The floor is by construction the one share of the push the source gate does not attenuate, so momentum briefly re-inflates exactly that share, a miniature of the directional gate’s re-inflation. The dose-response on the same exact buffer supports the reading: the floored push without momentum costs 0.2 pp (C8), with momentum 2.0 pp ($C8_M$), and the unfloored abrupt switch with momentum 14.1 pp (the same ungated exact-gradient rung, which is not seed-paired with $C8_M$ and is read as a magnitude only). $C8_M$ ’s area additionally counts in full under the end-referencing convention of Section 5.1.2, since it recovers above its pre-switch level. C8/ $C8_M$ are diagnostic probes: the full buffer defeats the limited-memory premise, and the practical scalar-gate configuration remains $C7_M$ (3.7 pp / 0.03 / 0.931) at 1/60th the memory.

	C7	$C7_M$	C8	$C8_M$
	(1k, $\mu=0$)	(1k, $\mu=0.9$)	(full, $\mu=0$)	(full, $\mu=0.9$)
gap_depth (pp)	13.9	3.7	0.2	2.0
gap_area_end	2.7	0.03	0.05	0.38
ACC	0.837	0.931	0.818	0.955

Table C.3: Curriculum $\times$ full buffer $\times$ momentum on rot-MNIST, isolating momentum’s two roles. Source variance reduction alone (C8) closes the gap; acceleration alone (the μ column) restores accuracy. The residual at C7 ($\mu=0$) is variance, not structure.

C.3 rot-MNIST: Buffer-Fidelity Diagnostics

The gradient diagnostics of Appendix B were logged on the vanilla-ER reference runs only. Each such step costs a full forward and backward pass over the whole past-task training set, which is why they are not carried elsewhere, and the stretched rungs of the ladder least of all (Appendix A.4). The rows below are therefore the directional and magnitude error that a 1,000-sample reservoir incurs against g_{true} at the standard step size, and they set the figure the ladder’s exact arm is defined against: a 60k buffer replays the deterministic empirical past-task gradient, so its cosine and magnitude ratio are 1.000 by construction.

Condition	$\bar{c}\bar{o}s$	$\bar{c}\bar{o}s_{\min}$	$\bar{\rho}$
D1 vanilla ER ($\mu=0$)	0.711 ± 0.035	0.086 ± 0.103	1.040 ± 0.039
D1 vanilla ER ($\mu=0.9$)	0.551 ± 0.033	0.135 ± 0.069	0.418 ± 0.014

Table C.4: Buffer-fidelity diagnostics for the vanilla-ER reference: $\bar{c}\bar{o}s$ and $\bar{c}\bar{o}s_{\min}$ are the mean and minimum of $\cos(g_{\text{replay}}, g_{\text{true}})$ over the window; $\bar{\rho}$ is the mean of $\|g_{\text{replay}}\|/\|g_{\text{true}}\|$.

C.4 rot-MNIST: Asymmetric PER (P-series)

Table C.5 gives the full metric set for the asymmetric-PER δ sweep on both buffer legs and at both momentum settings; the momentum-0 depth/area/ACC columns match Table 6.2 in the main text, and the momentum-0.9 rows are the source of the re-inflation numbers of Section 6.3. Momentum-0.9 runs cover the three strongest filters on the standard leg and $\delta \in \{0.3, 0.1\}$ on the full-buffer leg.

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end
<i>Standard (1k) buffer, momentum 0.0</i>								
$\delta=1.0$	0.869 ± 0.005	0.027 ± 0.018	0.792 ± 0.012	0.081 ± 0.009	0.733 ± 0.009	0.838 ± 0.006	9.0 ± 0.6	3.29 ± 0.65
$\delta=0.3$	0.873 ± 0.004	0.023 ± 0.017	0.807 ± 0.012	0.068 ± 0.009	0.736 ± 0.008	0.847 ± 0.005	7.4 ± 1.1	2.32 ± 0.59
$\delta=0.1$	0.876 ± 0.004	0.018 ± 0.016	0.822 ± 0.012	0.056 ± 0.009	0.739 ± 0.008	0.856 ± 0.005	5.9 ± 1.5	1.56 ± 0.65
$\delta=0.03$	0.880 ± 0.005	0.012 ± 0.015	0.834 ± 0.012	0.048 ± 0.005	0.741 ± 0.007	0.862 ± 0.005	4.8 ± 1.9	1.04 ± 0.58
<i>Standard (1k) buffer, momentum 0.9</i>								
$\delta=0.3$	0.929 ± 0.002	0.061 ± 0.004	0.851 ± 0.008	0.065 ± 0.004	0.813 ± 0.009	0.907 ± 0.005	10.4 ± 0.5	0.50 ± 0.17
$\delta=0.1$	0.928 ± 0.002	0.061 ± 0.006	0.861 ± 0.008	0.060 ± 0.005	0.815 ± 0.009	0.911 ± 0.005	9.4 ± 0.5	0.38 ± 0.19
$\delta=0.03$	0.929 ± 0.003	0.060 ± 0.009	0.871 ± 0.008	0.053 ± 0.007	0.816 ± 0.010	0.917 ± 0.005	8.5 ± 0.6	0.23 ± 0.12
<i>Full (60k) buffer, momentum 0.0</i>								
$\delta=1.0$	0.898 ± 0.002	-0.025 ± 0.012	0.841 ± 0.011	0.037 ± 0.011	0.734 ± 0.006	0.865 ± 0.006	4.0 ± 1.0	2.32 ± 1.38
$\delta=0.3$	0.900 ± 0.002	-0.029 ± 0.012	0.867 ± 0.002	0.026 ± 0.007	0.739 ± 0.005	0.878 ± 0.002	1.5 ± 1.1	1.01 ± 0.91
$\delta=0.1$	0.901 ± 0.002	-0.032 ± 0.013	0.860 ± 0.026	0.040 ± 0.028	0.741 ± 0.004	0.875 ± 0.013	2.2 ± 1.6	0.44 ± 0.38
$\delta=0.03$	0.903 ± 0.002	-0.035 ± 0.012	0.787 ± 0.094	0.118 ± 0.102	0.742 ± 0.004	0.838 ± 0.048	9.5 ± 8.2	0.43 ± 0.25
$\delta=0.03$ (CG 60)	0.903 ± 0.002	-0.035 ± 0.012	0.766 ± 0.113	0.126 ± 0.109	0.742 ± 0.004	0.828 ± 0.057	11.5 ± 10.1	0.47 ± 0.28
<i>Full (60k) buffer, momentum 0.9</i>								
$\delta=0.3$	0.964 ± 0.001	-0.012 ± 0.002	0.864 ± 0.009	0.052 ± 0.005	0.813 ± 0.008	0.913 ± 0.005	9.1 ± 0.8	1.91 ± 0.26
$\delta=0.1$	0.965 ± 0.001	-0.012 ± 0.003	0.880 ± 0.008	0.044 ± 0.004	0.814 ± 0.008	0.921 ± 0.005	7.6 ± 0.7	1.57 ± 0.25

Table C.5: rot-MNIST asymmetric-PER δ sweep, full metrics: both buffer legs at both momentum settings, plus the solver control (CG 60). The momentum-0.9 rows show the re-inflation of Section 6.3: depth climbs on every leg relative to its momentum-0 counterpart, while the full-buffer accuracies (≈ 0.965) are diagnostic values only, the full buffer defeating the limited-memory premise.

C.5 rot-MNIST: Symmetric vs. Asymmetric Preconditioning

Leaving the restoring gradient g_{rep} at full strength is the directional gate’s defining choice (Section 6.3). The alternative is to precondition the summed update, $d = \delta(F_{\text{rep}} + \delta I)^{-1}(g_{\text{cur}} + g_{\text{rep}})$, the symmetric variant that passes both gradients through the same metric. Because g_{rep} is the gradient of the replay loss, it lies in the top eigenspace of that loss’s own Fisher F_{rep} , so the symmetric filter shrinks the restoring gradient *harder* than the harmful push it is meant to brake and cannot preferentially protect the past task. Table C.6 sweeps the same δ on both variants, and they are mirror images. As the filter strengthens ($\delta \rightarrow 0$) the asymmetric gate’s forgetting falls (FORG $0.027 \rightarrow 0.012$) and its accuracy rises, while the symmetric gate’s forgetting *climbs* ($0.030 \rightarrow 0.066$) and its accuracy falls; every one of these four trends holds across all five seeds. Symmetric depth barely moves across the sweep ($5.7 \rightarrow 6.7$ pp) and at $\delta=1.0$ even sits below the asymmetric gate’s (5.7 vs. 9.0 pp), because filtering the whole update also damps the per-step *expression* of the dip. This is the area caveat of Section 5.1.2 made concrete: the symmetric gate’s end-referenced area collapses toward zero ($2.7 \rightarrow 0.01$) not because the dip recovers but because it stops recovering, settling into a permanent accommodation that the area excludes and the forgetting metric records. Read on depth or transient area the symmetric gate looks competitive; read on forgetting and final accuracy it degrades monotonically as its filter strengthens. Asymmetry is what turns the filter from a reshuffling of the damage into a net removal of it.

δ	Asymmetric (g_{rep} free)				Symmetric (joint)			
	depth (pp)	area	FORG	ACC	depth (pp)	area	FORG	ACC
1.0	9.0	3.29	0.027	0.869	5.7	2.69	0.030	0.872
0.3	7.4	2.32	0.023	0.873	5.9	1.65	0.042	0.867
0.1	5.9	1.56	0.018	0.876	6.2	0.34	0.057	0.861
0.03	4.8	1.04	0.012	0.880	6.7	0.01	0.066	0.857

Table C.6: Asymmetric versus symmetric preconditioning on the standard 1k buffer at momentum 0, five seeds, sweeping the shared damping δ . Both variants filter through the replay Fisher F_{rep} ; they differ only in whether the metric is applied to the new-task push alone (asymmetric, g_{rep} left free) or to the summed update (symmetric). As $\delta \rightarrow 0$ the asymmetric gate improves on every axis while the symmetric gate degrades on every axis; symmetric depth stays flat because the filter suppresses the per-step expression of the dip even as the underlying forgetting grows, and the symmetric end-referenced area collapses toward zero for the same reason: the past task stops recovering, so the loss registers in FORG rather than in the transient area (Section 5.1.2).

C.6 CIFAR-10: Headline Block

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end
G1 vanilla BN	0.723 ± 0.008	0.010 ± 0.019	0.644 ± 0.023	0.127 ± 0.015	0.484 ± 0.020	0.667 ± 0.016	11.8 ± 2.5	17.5 ± 4.6
G2 adaptive BN	0.722 ± 0.011	0.006 ± 0.016	0.713 ± 0.017	0.070 ± 0.010	0.469 ± 0.016	0.698 ± 0.013	4.8 ± 0.7	3.6 ± 2.2
G3 asym. PER BN	0.715 ± 0.011	-0.008 ± 0.018	0.698 ± 0.023	0.124 ± 0.022	0.460 ± 0.025	0.682 ± 0.023	5.9 ± 4.3	3.3 ± 3.0
G4 vanilla GN	0.730 ± 0.011	-0.023 ± 0.030	0.309 ± 0.111	0.294 ± 0.097	0.415 ± 0.058	0.507 ± 0.054	41.2 ± 13.5	34.9 ± 23.7
G5 adaptive GN	0.731 ± 0.020	-0.026 ± 0.026	0.670 ± 0.055	0.095 ± 0.033	0.340 ± 0.052	0.687 ± 0.037	5.2 ± 2.1	1.7 ± 0.9
G6 asym. PER GN	0.741 ± 0.009	-0.035 ± 0.018	0.673 ± 0.023	0.080 ± 0.009	0.330 ± 0.024	0.696 ± 0.015	5.6 ± 1.1	4.8 ± 3.8

Table C.7: CIFAR-10 headline block, full metrics (ResNet-18, momentum 0.9, buffer 5,000, five seeds). “adaptive” is the practical curriculum with $\lambda_{\min} = 0.20$; “asym. PER” runs at $\delta = 0.1$.

The gates’ residual batch-norm dip. The small gap the gates keep under batch norm is a genuine boundary transient: in Figure 6.6(a) both gated curves dip together over the first steps after the switch, by roughly the tabulated 5 pp, and recover to their plateau within a few tens of steps. A reading consistent with this pattern is the normalisation pathway of Section 5.2.5. BatchNorm’s running statistics update on every forward pass, whatever the gate does to the gradient, so the first corrupted batches shift them away from the clean task and no gate on the update can hold that back; the clean replay samples in every batch pull the statistics back, which matches the fast recovery. The group-norm contrast supports the reading: with no cross-batch statistics to disturb, the curriculum holds T_0 flat through the same boundary (Figure 6.6(b)) and its transient area falls to roughly half its batch-norm value (1.7 vs. 3.6). This is a reading rather than an established mechanism; a control that freezes the running statistics across the switch would isolate it.

C.7 CIFAR-10: Generalisation Block

Each shift pairs the practical curriculum (“curric.”: adaptive $\lambda_{\min} = 0.20$, buffer 5,000) against a vanilla-ER control on the same shift, at the indicated normalisation, over three seeds.

Condition	ACC	FORG	min-ACC	WC-ACC	Depth	Area end
<i>BatchNorm</i>						
shot noise (vanilla)	0.724 ± 0.011	0.017 ± 0.010	0.633 ± 0.017	0.663 ± 0.014	13.1 ± 3.0	21.2 ± 3.9
shot noise (curric.)	0.723 ± 0.014	0.010 ± 0.010	0.708 ± 0.032	0.697 ± 0.016	5.6 ± 1.0	5.8 ± 1.5
contrast (vanilla)	0.671 ± 0.026	-0.008 ± 0.006	0.610 ± 0.063	0.586 ± 0.048	15.4 ± 3.9	33.4 ± 4.5
contrast (curric.)	0.671 ± 0.044	-0.012 ± 0.007	0.727 ± 0.034	0.643 ± 0.048	3.7 ± 1.1	2.2 ± 0.9
3-task (vanilla)	0.740 ± 0.006	-0.011 ± 0.013	0.670 ± 0.015	0.691 ± 0.010	3.3 ± 0.4	6.0 ± 0.7
3-task (curric.)	0.745 ± 0.010	-0.021 ± 0.008	0.677 ± 0.025	0.696 ± 0.017	6.9 ± 2.7	3.9 ± 0.1
<i>GroupNorm</i>						
shot noise (vanilla)	0.718 ± 0.021	-0.036 ± 0.036	0.244 ± 0.075	0.469 ± 0.032	46.3 ± 12.0	54.4 ± 40.8
shot noise (curric.)	0.705 ± 0.061	-0.028 ± 0.012	0.638 ± 0.108	0.656 ± 0.087	7.3 ± 5.9	2.4 ± 1.1
contrast (vanilla)	0.777 ± 0.022	-0.067 ± 0.021	0.641 ± 0.061	0.710 ± 0.040	8.4 ± 4.4	3.3 ± 0.6
contrast (curric.)	0.778 ± 0.032	-0.070 ± 0.013	0.668 ± 0.077	0.724 ± 0.054	3.9 ± 3.1	1.2 ± 0.8
3-task (vanilla)	0.722 ± 0.020	-0.023 ± 0.015	0.468 ± 0.043	0.550 ± 0.023	4.4 ± 0.4	10.3 ± 3.6
3-task (curric.)	0.721 ± 0.019	-0.025 ± 0.020	0.421 ± 0.114	0.517 ± 0.070	32.8 ± 14.5	32.0 ± 19.5

Table C.8: CIFAR-10 generalisation block, full metrics. The two-task novel corruptions (shot noise, contrast) favour the curriculum under both normalisations; the three-task group-norm sequence is the outlier analysed below.

The three-task group-norm failure. The three-task sequence is the one place the curriculum underperforms its control (depth 32.8 ± 14.5 pp against vanilla’s 4.4 ± 0.4 , Table C.8; Figure C.1), and the logged gradient-norm traces locate the failure in the conditioning of the feedback signal. The adaptive ratio $r = \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ carries information through the boundary ordering of Section 5.2.2: at a well-separated switch the new-task gradient dwarfs the replay gradient, so r starts near zero and its rise tracks the replay gradient reasserting itself. The first boundary of this sequence behaves that way (clean $\rightarrow$ Gaussian: first-step new-task loss 1.9–2.1, $r \approx 0.2$), and the schedule holds every past task flat through it. The second boundary never establishes the ordering: Gaussian and shot noise are near-identical, so the incoming model already fits the new task (T_2 accuracy 0.67–0.71 before its first step, new-task loss 0.5–0.7), leaving $\|g_{\text{new}}\|$ small, while $\|g_{\text{replay}}\|$ is small because the model is converged on the replayed tasks. The ratio becomes a quotient of two near-zero norms: under group norm it reads 0.5–1.3 from the first step and fluctuates step to step, so the EMA drives λ from the floor to ≈ 0.7 –0.8 within fifty steps and back to $\lambda_{\min}$ within a few hundred. The gate is moved by noise, not by damage.

What follows, in all three seeds, is not a stability gap but an optimisation instability. Within the first forty steps the replay loss itself diverges (0.2 – $0.4 \rightarrow 0.8$ – 2.1) and every task falls together, *including the one being trained* (T_2 : $0.71 \rightarrow 0.29$, $0.69 \rightarrow 0.30$, $0.67 \rightarrow 0.51$ across seeds), before the run recovers; the per-seed depth scales with the divergence, which is where the ± 14.5 pp spread comes from. Because all tasks crash together,

the depth and area of the three-task group-norm rows record this instability, not the past-task transient the rest of the study measures. Two controls isolate the trigger. The vanilla run crosses the same boundary with a flat replay loss and 4.4 pp of depth, so the near-identical boundary is harmless on its own; the batch-norm run executes the identical schedule through the same boundary, with the same ill-conditioned early ratio, without instability (Figure C.1), so the noise-driven λ is not sufficient either. The failure needs both the unconditioned signal and the group-norm backbone’s boundary sensitivity, consistent with the far deeper group-norm transients vanilla ER shows on the same corruption family (shot noise: 46.3 vs 13.1 pp, Table C.8). We therefore read the episode as a reliability boundary of the gradient-norm-ratio signal rather than of the gate principle: the ratio loses its meaning when both norms become small, and the design admits direct guards: freezing λ when both norms fall below a reference scale, or reading a damage signal that does not vanish at a quiet boundary, such as the replay-loss excess over its pre-switch level.

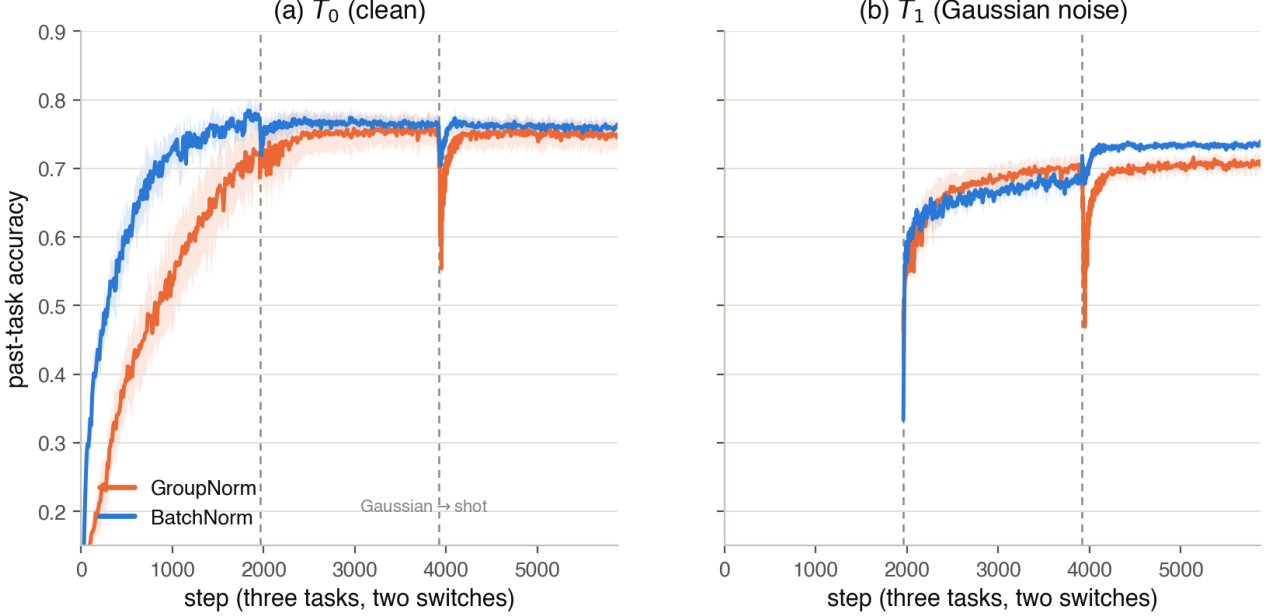


Figure C.1: Per-step past-task accuracy across the three-task CIFAR-10 generalisation sequence (none $\rightarrow$ Gaussian $\rightarrow$ shot, ResNet-18, practical curriculum, three seeds with $\pm$ std bands), comparing the group-norm run (orange) against its batch-norm counterpart (blue). The two switches (dashed) fall at $\sim 2k$ and $\sim 4k$ steps. The first boundary (none $\rightarrow$ Gaussian) barely perturbs the past tasks in either norm. The second (Gaussian $\rightarrow$ shot, two near-identical corruptions) drives a deep transient in T_0 (a) and T_1 (b) under group norm only, while the batch-norm run holds flat through both switches. The dip is thus localised entirely to the second transition and to group norm, the adaptive-ratio reliability boundary analysed above.

C.8 CIFAR-10: Momentum Cross and Compute Cost

This subsection collects the three deep-backbone checks summarised in Section 6.5: the momentum cross (Table C.9), the generalisation block (Table C.8 above), and the compute cost (Table C.10).

Momentum cross. The cross repeats the composability test of Section 5.2.3 on the ResNet, pairing momentum-0 runs at group norm against the momentum-0.9 rows, and it turns out to be only weakly informative. At momentum 0 the ResNet under-converges the past task in ten epochs (vanilla min-ACC 0.11, ACC 0.65), so every momentum-0 depth sits on a degraded baseline. On that baseline both gates improve with momentum by similar margins (curriculum 28.6 $\rightarrow$ 5.2 pp, asymmetric PER 22.6 $\rightarrow$ 5.6 pp), so the cross cannot separate the predicted source-attenuation composition from a general convergence gain, and the prediction that momentum re-inflates the directional gate does not appear; the vanilla null (40.5 $\rightarrow$ 41.2 pp) is directionally consistent with the account but rests on the same degraded baseline. We therefore draw no mechanism conclusion from this cross in either direction: the momentum signatures rest on the rot-MNIST results, where the MLP isolates them cleanly (Sections 6.2 and 6.3), and a clean deep-backbone test would need a converged momentum-0 baseline (Section 7.4).

Method (GroupNorm)	depth, $\mu=0$	depth, $\mu=0.9$
Vanilla ER	40.5 ± 4.0	41.2 ± 13.5
Curriculum ($\lambda_{\min}=0.20$)	28.6 ± 8.8	5.2 ± 2.1
Asym. PER ($\delta=0.1$)	22.6 ± 9.7	5.6 ± 1.1

Table C.9: CIFAR-10 momentum cross (group norm, buffer 5,000, five seeds), gap depth in pp. All momentum-0 cells sit on an under-converged baseline (vanilla min-ACC 0.11), so the cross is read as uninformative for the momentum signatures rather than as partial confirmation (see text).

Compute cost. Cost is measured with a dedicated benchmark: one full two-task run per gate in its headline configuration (curriculum: adaptive schedule, $\lambda_{\min}=0.20$; asymmetric PER: $\delta=0.1$, 25 warm-started CG iterations; momentum 0.9, group norm and buffer 5,000 on CIFAR), five seeds each on a single NVIDIA RTX PRO 6000, with per-step stability evaluation disabled so the numbers reflect method compute alone. Wall-clock and CPU time come from `getrusage`; GPU utilisation, power, and memory are sampled at 1 Hz with `nvidia-smi`, and energy is the integrated power. On the ResNet-18 the split is stark (Table C.10). The scalar gate adds one gradient-norm ratio and an EMA update to a standard ER step, and its run is not even compute-bound: the GPU sits half-idle at 49% utilisation and 259 W. The directional gate’s 25 conjugate-gradient Fisher-vector products per step, each on the order of a backward pass, saturate the same card (96%, 367 W) for $16\times$ as long, and since energy is power times time the two factors compound to a $24\times$ energy gap (13 vs. 320 Wh per run). Memory is not the price: peak GPU memory differs by 11%. On the rot-MNIST MLP both gates finish in under half a minute at near-idle utilisation (the wall-clock is fixed startup, not method compute), so no meaningful ratio exists there and the cost question is decided entirely on the deep backbone. The ratios are the transferable claim; the absolute numbers are specific to this card, and because $\sim 15\text{--}30$ s of each run is startup, the training-only ratio is if anything slightly above $16\times$.

Dataset	Gate	wall [s]	util [%]	power [W]	energy [Wh]	mem [MB]
CIFAR-10	Curriculum (scalar)	198 ± 7	49	259	13	4,203
(ResNet-18)	Asym. PER ($\delta=0.1$)	$3,157 \pm 11$	96	367	320	4,679
rot-MNIST	Curriculum (scalar)	25 ± 1	<1	—	0.4	749
(MLP)	Asym. PER ($\delta=0.1$)	29 ± 1	8	—	0.5	769

Table C.10: Compute cost of one full two-task run (mean over five seeds, $\pm$ sd on wall-clock; single RTX PRO 6000; per-step evaluation disabled): wall-clock, mean GPU utilisation and power, integrated GPU energy, and peak GPU memory. On the ResNet-18 the directional gate costs $\sim 16\times$ the wall-clock and $\sim 24\times$ the energy of the scalar gate at similar peak memory. The MLP rows are dominated by fixed startup on a near-idle GPU, so their power draw is idle draw (omitted) and no ratio is meaningful.

C.9 Long-Sequence rot-MNIST (L-series)

Condition	ACC	FORG	min-ACC	WF ₁₀₀	WP ₁₀₀	WC-ACC	Depth	Area end
<i>Momentum 0.0</i>								
L1 vanilla ER	0.861 ± 0.004	0.060 ± 0.004	0.670 ± 0.055	0.207 ± 0.053	0.443 ± 0.011	0.722 ± 0.043	21.0 ± 5.8	3.14 ± 0.77
L2 curriculum $\lambda_{\min}=0.20$	0.860 ± 0.002	0.031 ± 0.005	0.828 ± 0.007	0.073 ± 0.005	0.400 ± 0.003	0.840 ± 0.005	4.1 ± 0.8	1.62 ± 0.42
L3 asym. PER $\delta=0.1$	0.870 ± 0.003	0.053 ± 0.006	0.843 ± 0.008	0.038 ± 0.007	0.381 ± 0.003	0.862 ± 0.007	4.7 ± 1.2	0.38 ± 0.24
<i>Momentum 0.9</i>								
L1 vanilla ER	0.889 ± 0.007	0.096 ± 0.008	0.845 ± 0.007	0.065 ± 0.008	0.317 ± 0.006	0.870 ± 0.006	8.9 ± 1.7	1.68 ± 0.69
L2 curriculum $\lambda_{\min}=0.20$	0.897 ± 0.002	0.071 ± 0.003	0.871 ± 0.005	0.041 ± 0.004	0.364 ± 0.005	0.888 ± 0.004	5.4 ± 0.6	0.60 ± 0.14
L3 asym. PER $\delta=0.1$	0.884 ± 0.004	0.104 ± 0.004	0.849 ± 0.006	0.054 ± 0.004	0.325 ± 0.005	0.873 ± 0.005	7.5 ± 1.5	0.80 ± 0.63

Table C.11: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive transitions, five seeds), the L-series: both gates in their practical configurations against vanilla ER, at both momentum settings. Depth is the worst single-boundary drop and area the end-referenced transient, aggregated as in the two-task blocks. At momentum 0 both gates cut the depth from 21 pp to ≈ 4 pp; under momentum the curriculum keeps the best cumulative retention while the re-inflated directional gate falls slightly behind vanilla on FORG (0.104 vs. 0.096, all five seeds).

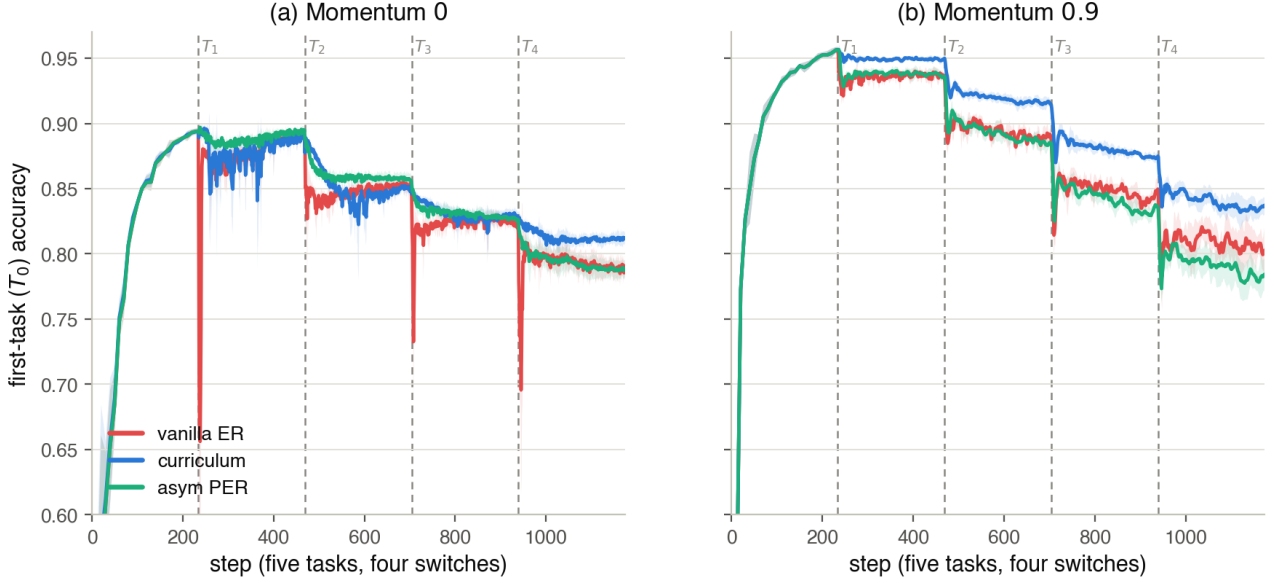


Figure C.2: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive switches, five seeds with $\pm$ std bands), the L-series: first-task T_0 accuracy across the whole sequence for vanilla ER (red), the practical curriculum $\lambda_{\min}=0.20$ (blue) and asymmetric PER $\delta=0.1$ (aqua); dashed lines mark the four switches, and the two panels are the two momentum settings. (a) Momentum 0: both gates hold T_0 almost flat through every boundary, asymmetric PER the smoothest, while vanilla ER drops sharply at each switch. (b) Momentum 0.9: PER’s boundary transients return as momentum re-inflates its per-step filter (Section 6.3); the curriculum keeps the boundaries smoothest and holds the highest plateau (Table C.11).

D Derivation of the Trajectory Bound

This appendix derives the two results used in Section 3.1: the replay loss is monotone along the reference path (Proposition 1), and a gradient-flow iterate under a linear ramp tracks that path with $O(1/N)$ lag (Proposition 2), which together give the transient bound of Equation (3.2).

Setup. For $\lambda \in [0, 1]$ define the curriculum objective

$$L_\lambda(\theta) = L_{\text{replay}}(\theta) + \lambda L_{\text{current}}(\theta), \quad H_\lambda(\theta) = H_{\text{replay}}(\theta) + \lambda H_{\text{current}}(\theta), \quad (\text{D.1})$$

with $H_{\text{replay}} = \nabla^2 L_{\text{replay}}$ and $H_{\text{current}} = \nabla^2 L_{\text{current}}$. We make two assumptions.

- (A1) L_{replay} and L_{current} are C^2 in a neighbourhood of the path considered below.
- (A2) There is a C^1 branch of critical points $\lambda \mapsto \Theta^*(\lambda)$ of L_λ with $\Theta^*(0) = \theta_0^*$ and $H_\lambda(\Theta^*(\lambda)) \succ 0$ for all $\lambda \in [0, 1]$.

Assumption (A2) asks for a tracked branch of strict local minima, not global convexity. Near $\lambda = 0$ it holds automatically: θ_0^* is a non-degenerate local minimum of L_{replay} , so $H_0(\theta_0^*) \succ 0$ and the implicit function theorem yields a unique C^1 branch locally. Its persistence is controlled by Weyl’s inequality, $\mu_{\min}(H_\lambda) \geq \mu_{\min}(H_{\text{replay}}) + \lambda \mu_{\min}(H_{\text{current}})$, with $\mu_{\min}$ the smallest eigenvalue: positive-definiteness survives until the new task’s negative curvature outweighs the old task’s, and holds for all λ whenever $H_{\text{current}} \geq 0$ along the branch. On a deep non-convex backbone (A2) is a local modelling assumption, which is why Section 3.1 presents the bound as motivation rather than a guarantee.

Proposition 1 (Monotonicity of the replay loss). *Under (A1)–(A2),*

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) = \lambda \nabla L_{\text{current}}^\top H_\lambda^{-1} \nabla L_{\text{current}} \geq 0 \quad \text{for all } \lambda \in [0, 1],$$

where $\nabla L_{\text{current}}$ and H_λ are evaluated at $\Theta^*(\lambda)$. In particular $L_{\text{replay}}(\Theta^*(\lambda)) \leq L_{\text{replay}}(\theta_{\text{joint}}^*)$ for all λ , with $\theta_{\text{joint}}^* = \Theta^*(1)$.

Proof. The branch satisfies the first-order condition

$$\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0. \quad (\text{D.2})$$

Differentiating (D.2) in λ and collecting terms gives $H_\lambda \frac{d\Theta^*}{d\lambda} + \nabla L_{\text{current}} = 0$, hence

$$\frac{d\Theta^*}{d\lambda} = -H_\lambda^{-1} \nabla L_{\text{current}}. \quad (\text{D.3})$$

By the chain rule and (D.2),

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) = \nabla L_{\text{replay}}^\top \frac{d\Theta^*}{d\lambda} = (-\lambda \nabla L_{\text{current}})^\top (-H_\lambda^{-1} \nabla L_{\text{current}}) = \lambda \nabla L_{\text{current}}^\top H_\lambda^{-1} \nabla L_{\text{current}},$$

which is non-negative because $\lambda \geq 0$ and $H_\lambda^{-1} \succ 0$ by (A2). Integrating from λ to 1 gives the final statement. $\square$

Proposition 2 ($O(1/N)$ tracking lag). *Let $\theta(t)$ follow the gradient flow $\dot{\theta} = -\nabla L_{\lambda(t)}(\theta)$ under the linear ramp $\lambda(t) = t/N$, with $\theta(0) = \theta_0^*$, and let $e(t) = \theta(t) - \Theta^*(\lambda(t))$ denote the tracking error. Suppose, in addition to (A1)–(A2), that along the branch $\mu_{\min}(H_\lambda) \geq \mu > 0$ and $\|\nabla L_{\text{current}}\| \leq g$. Then, to first order in e , the error settles at*

$$\|e\| \leq \frac{1}{N} \|H_\lambda^{-1}\| \left\| \frac{d\Theta^*}{d\lambda} \right\| \leq \frac{g}{\mu^2 N} = O\left(\frac{1}{N}\right).$$

Proof. Differentiating the error and inserting the flow and the chain rule for the moving target,

$$\dot{e} = -\nabla L_{\lambda(t)}(\theta(t)) - \frac{d\Theta^*}{d\lambda} \dot{\lambda}, \quad \dot{\lambda} = \frac{1}{N}.$$

Linearising the gradient about the minimiser, $\nabla L_\lambda(\theta) = H_\lambda e + O(\|e\|^2)$ since $\nabla L_\lambda(\Theta^*) = 0$, yields the error dynamics

$$\dot{e} \approx -H_\lambda e - \frac{1}{N} \frac{d\Theta^*}{d\lambda}.$$

Because $H_\lambda \succeq \mu I$, these dynamics contract toward the quasi-static equilibrium $e^\dagger = -\frac{1}{N} H_\lambda^{-1} \frac{d\Theta^*}{d\lambda}$; bounding $\|H_\lambda^{-1}\| \leq 1/\mu$ and, by (D.3), $\|d\Theta^*/d\lambda\| \leq g/\mu$ gives the stated bound. The argument is a first-order, continuous-time approximation: it captures how the lag scales with the ramp length N , not an exact discrete-time constant. $\square$

The transient bound. Expanding the replay loss about the reference point, $L_{\text{replay}}(\theta(t)) = L_{\text{replay}}(\Theta^*) + \nabla L_{\text{replay}}(\Theta^*)^\top e + O(\|e\|^2)$, Proposition 1 bounds the first term by $L_{\text{replay}}(\theta_{\text{joint}}^*)$ and Proposition 2 bounds the correction by $O(1/N)$ (the gradient norm $\|\nabla L_{\text{replay}}(\Theta^*)\| = \lambda \|\nabla L_{\text{current}}\| \leq g$ is bounded along the branch). Hence

$$\max_t L_{\text{replay}}(\theta(t)) \leq L_{\text{replay}}(\theta_{\text{joint}}^*) + O\left(\frac{1}{N}\right): \quad (\text{D.4})$$

the permanent price is the first term, and the transient contribution of the tracking lag shrinks as the ramp lengthens, which is Equation (3.2).